

BEYOND INDIVIDUAL CLAIMS: PROBABILISTIC LOGICAL COHERENCE FOR LONG-FORM LLM EVALUATION

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ABSTRACT

Long-form factuality assessment decomposes a response into atomic claims and verifies each against retrieved evidence. The paradigm is effective and now standard, but it embeds an assumption — that the truth of each claim can be settled in isolation — which discards the structure that makes prose an argument. A response can therefore pass per-claim verification while contradicting itself, and two responses built from *identical* claims can differ in coherence because of the order in which they assert them. We formalize *logical coherence* as a property of the joint distribution over claim truth values induced by a Markov random field whose pairwise factors are the relations the response itself draws. The relation vocabulary is a two-level taxonomy: twelve interpretable discourse senses, grounded in PDTB 3.0 and RST, that compile through a deterministic map onto five inferential couplings, each of which fixes a factor table by naming the joint assignment it penalizes. We characterize the factor tables coherence requires by a row-stochasticity property and show that the tables used for claim–evidence factuality are wrong here, collapsing the measure’s dynamic range. We then define four candidate coherence readouts over the model — the mean posterior marginal, a two-term consistency probability, a reified coherence node, and a normalized log-partition — give the exact auxiliary-variable construction each requires, and evaluate all four against five stated criteria by exact inference. Only two are monotone in conflict resolution, and the two that fail do so for one shared and instructive reason: they measure whether a conflict is *active* rather than what the reader should *believe*. A third natural candidate is worse still, since a score asking whether asserted relations are honoured is maximized by asserting nothing. Claim priors come from an existing factuality pipeline, making coherence and factuality two stages over one decomposition rather than competing scores. Empirically, on a corpus of response families whose relative coherence ordering is fixed by construction, the readouts recover 78.6% of the declared orderings from gold relation graphs and 53.6% from graphs mined out of the response text; since the two conditions differ only in the relation graph, the gap localizes the open problem in relation extraction rather than in the probabilistic model.

1 INTRODUCTION

The dominant paradigm for assessing the factuality of long-form generation is decompose-then-verify: split a response into atomic claims, check each against retrieved evidence, and aggregate the verdicts (Min et al., 2023; Wei et al., 2024; Song et al., 2024). It is effective, it is now standard, and it embeds an assumption. Treating each claim as independently checkable discards precisely the structure that distinguishes an *argument* from a *list*.

Consider two failure modes that per-claim verification cannot see. A report may assert both that no passengers were harmed in an incident and that a wire service reported three deaths. One of these claims is true and one is false, and per-claim verification will say so — but it will not register the more basic defect, that the response contradicts *itself*, asserting two things that cannot both hold. Separately, the inferential links a response draws may fail even when their endpoints are individually true: a chain of accurate statements joined by a spurious “therefore” is not a sound argument, and

every atom in it verifies. Xie et al. (2026) quantify how often this matters: on logic-related samples, SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of cases respectively.

Factuality and coherence are orthogonal axes. A response can be a bag of true facts that is incoherent, and it can be internally impeccable while being entirely false. The sharpest demonstration in our fixtures is a pair of summaries of one appellate judgment, built from the *same* set of claims and differing only in whether self-serving assertions are stated before or after the findings that qualify them. Every atom is identical, so every per-atom check returns an identical verdict; yet one summary is a faithful account and the other is not, and our measure separates them (0.95 against 0.90). No amount of grounding individual claims against evidence can make that distinction, because the distinction is not about the claims.

Coherence is a statement about a joint distribution. This paper measures the second axis, and the central modelling commitment is that the right object is a joint distribution rather than a tally. Given what a response asserts and how it links its assertions, we ask: how much of what it asserts does the argument, taken as a whole, uphold? That question is answered by a *marginal* in a probabilistic graphical model whose variables are claim truth values and whose factors are the asserted relations — not by counting how many rules a configuration violates. The distinction is not stylistic. A conflict between two claims should propagate: it should depress belief in both endpoints, and through them in whatever those endpoints support. Counting violated rules cannot propagate, and as we show in §5, scoring rule satisfaction directly produces a measure that a response can maximize by asserting no relations at all.

Our contributions are as follows.

- **A coherence MRF and a formal two-level relation taxonomy** (§4). Twelve discourse senses compile through a deterministic map onto five inferential couplings, each defined by the joint assignment it penalizes, which fixes its factor table. Separating the interpretable layer from the inferential one lets the annotation vocabulary be rich while the probabilistic model stays a closed three-parameter pairwise family that no relation type can extend.
- **A characterization of why factuality’s factor tables are wrong for coherence** (§4.3). The tables that are correct for claim–evidence factuality charge a premise for being a premise. We characterize the tables coherence requires by a row-stochasticity property, and show the alternative collapses the measure: the root of an entailment chain falls to $\mathbb{P}(a_1=1) = 0.19$ and the mean marginal moves only from 0.490 to 0.498 when every conflict edge is deleted, against 0.607 to 0.620 for the correct tables.
- **Four readouts, their exact constructions, and a selection argument** (§5). We give the deterministic auxiliary-variable constructions that make two of them tractable without a $2^{|\mathcal{R}|}$ table, evaluate all four by exact inference across a coherence-ordered family, and establish why the mean posterior marginal is the one to report: it alone satisfies all five criteria, and the two failures share a diagnosable cause (Prop. 3).
- **Two-stage composition with factuality** (§6), taking posterior marginals from an existing factuality pipeline as the coherence model’s priors, so one decomposition and one inference stack serve both axes and either can be read alone.
- **Measurements** (§8) that separate relation-extraction error from scoring error by holding the readouts fixed and swapping only the relation graph, plus a worked incoherent/coherent pair with exact values (§9) and a Markov-logic correspondence establishing what the pairwise encoding does and does not buy (Appendix C).

2 RELATED WORK

Atomic factuality. FActScore (Min et al., 2023) introduced per-atom precision against a knowledge source, decomposing a biography into atomic facts and reporting the fraction supported. SAFE (Wei et al., 2024) added search-augmented verification with a multi-step reasoning agent per claim and the $F_1@K$ metric that trades precision against a recall target. VeriScore (Song et al., 2024) restricted attention to *verifiable* claims, on the argument that many decomposed atoms are not checkable in

principle. FactVerify (Bayat et al., 2025) reasons over retrieved evidence with an undecided verdict class. All four aggregate independent per-claim verdicts, and all four are therefore blind to inter-claim structure by construction.

FactReasoner (Marinescu et al., 2025) replaced verdict counting with probabilistic inference: it builds a factor graph over claim and evidence variables with pairwise factors derived from natural-language-inference relations, and reports posterior marginals rather than a tally. This is the closest antecedent of our model, and we reuse its decomposition, its relation extractor and its inference stack (§3), and take its marginals as our priors (§6). The question it asks is nonetheless different: its factors run between a claim and the *evidence* retrieved for it, so its graph has no claim–claim edges and its marginals cannot express internal consistency. Coherence is a different question asked of the same decomposition.

The logic gap. Xie et al. (2026) states the problem we address: decompose-then-verify pipelines verify atoms individually and overlook the logical dependencies among them, so a text combining true facts through a false dependency is misjudged as factual. Their remedy classifies dependencies as a labelling task over claim pairs. Our departure is to treat the dependencies as *factors in a joint distribution*, which is what allows a local conflict to propagate to distant claims and what makes the resulting score a functional of the whole graph rather than a per-edge average.

Discourse coherence. Entity-grid models (Barzilay & Lapata, 2008) score coherence from the distribution of entity mentions across sentences, and their neural successors (Li & Jurafsky, 2017) learn the same signal with distributed representations. Both are trained and evaluated against *synthetic perturbations* — sentence shuffling and, in later work, contradiction injection — a methodology we extend in §7 by making the perturbation an operator on a declared relation graph rather than on raw text, so its intended effect on the score is known per item. DiscoScore (Zhao et al., 2023) scores generated text using BERT-based discourse representations and correlates well with human coherence judgements. What these approaches share is that coherence is read off surface statistics or learned representations; none exposes an inspectable relation graph, and none yields per-claim attributions. Our relation inventory takes its senses from PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988), but treats them as an interpretable layer that compiles to a smaller inferential vocabulary — which is what makes discourse senses usable as factor types at all.

Consistency checking and LLM judges. SelfCheckGPT (Manakul et al., 2023) detects hallucination by sampling several responses and running pairwise NLI between them; its NLI variant is the natural ablation isolating what global inference adds over purely local contradiction detection, and we propose it as such in §7.1. G-Eval (Liu et al., 2023) is the practical incumbent: an LLM judge scoring a coherence dimension directly from a rubric. Judges of this kind are strong on aggregate correlation and weak on exactly what a graphical model provides — attribution to specific claims, and a guarantee that a meaning-preserving edit cannot change the score.

Relational probabilistic models. Markov logic (Richardson & Domingos, 2006) is the natural alternative encoding, replacing factor tables with weighted first-order rules. Appendix C works out the correspondence and finds it instructive in both directions: the obvious one-clause-per-relation encoding is *not* equivalent to our model and is degenerate at its mode, while a three-clause expansion reproduces it exactly on the three-coupling fragment. We retain the pairwise MRF for the measure because its factor family is closed, and retain the Markov logic view because the beyond-pairwise rules that would address our main modelling limitation are expressible there.

3 STAGE ONE: PROBABILITIES FOR INDIVIDUAL CLAIMS

The coherence model consumes claim-level probabilities and reuses the inference machinery that produces them, so we describe that stage before defining the model itself. It is FactReasoner (Marinescu et al., 2025); the account here is a summary sufficient to make the composition of §6 precise, and it also fixes notation and states the estimator whose label-span construction the relation miner reuses in §4.5.

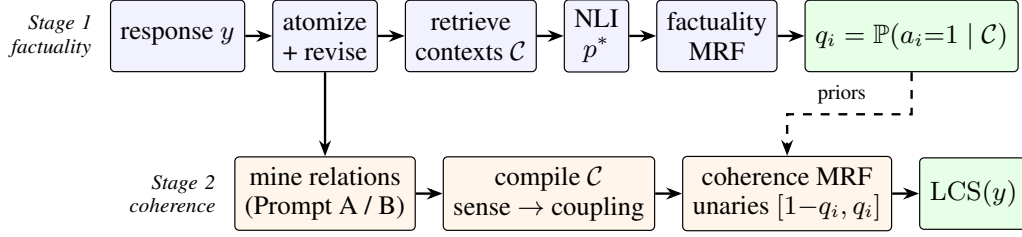


Figure 1: The two-stage pipeline. Stage 1 (§3) is FactReasoner (Marinescu et al., 2025): it grounds each atom against retrieved evidence and returns a posterior marginal q_i . Stage 2 (§4) builds a second MRF over the atoms *alone*, takes q_i as its unary priors, and adds one pairwise factor per mined atom–atom relation. Atomization is shared, so the response is decomposed once. Setting $q_i \equiv \frac{1}{2}$ recovers a pure coherence score that consults no external evidence.

3.1 DECOMPOSITION, RETRIEVAL, AND RELATION EXTRACTION

A response y is decomposed by an LLM *atomizer* into atomic claims $\mathcal{A} = \{a_1, \dots, a_n\}$, each a single independently verifiable proposition. A *reviser* then decontextualizes each one, resolving pronouns and anaphora so that the claim is checkable outside its original sentence; near-duplicates are collapsed. For each claim a retriever returns supporting passages, which are deduplicated into an evidence set $\mathcal{C} = \{c_1, \dots, c_m\}$.

An LLM-based NLI extractor then labels admitted pairs. For a claim–evidence pair the label is drawn from {entailment, contradiction, neutral}; for an evidence–evidence pair both directions are scored and reconciled, with entailment in both directions relabelled *equivalence*. Neutral verdicts are dropped rather than encoded: a claim about which the evidence is silent contributes no factor and is left at its prior, which is what makes the factor set of §4 contain only the relations someone actually asserted.

3.2 READING A PROBABILITY OFF THE GENERATION

Each retained relation carries a probability p^* . Rather than asking the model to verbalize a confidence, we read it from the distribution over the tokens it actually emitted. The extractor is prompted to end with a bracketed label; let T be the set of generated tokens overlapping the span of that label in the decoded text. Then

$$p^* = \exp\left(\frac{1}{|T|} \sum_{t \in T} \log p_t\right), \quad (1)$$

the per-token geometric-mean probability of the label the model produced. Three details of this construction are load-bearing, and we record them because each was a source of silent error.

Span alignment. The probability is measured over exactly the span the label *parser* reports, reconstructing the decoded string from the token list and tracking each token’s character offsets. Measuring the same span the label came from makes the label and its probability structurally unable to disagree. Requiring standalone bracket-delimiter tokens instead does not work: tokenizers routinely fuse [,] and " into subwords with adjacent characters, so a delimiter-based rule silently fails on some backends and not others.

Token-array conventions. An earlier version of this code stripped the final entry of the logprob array as an end-of-sequence token. OpenAI-compatible and vLLM logprob arrays contain only emitted content tokens — the stop is signalled out of band — so dropping the last entry deleted a real content token, and for a bracketed label that token is the one closing the label whose confidence is being measured.

Degenerate cases. When the logprob array is empty, no label span is found, or no token aligns to it, the estimator returns 0.5 rather than 0.0. Returning zero would be read downstream as a *degenerate*, *impossible* relation for a label the model in fact generated, which is a stronger claim than “we could not measure the confidence”.

Backends without logprobs. Some serving stacks expose no token logprobs at all. There p^* instead comes from SIMBA-UQ (Bhattacharjya et al., 2025), which samples the extractor repeatedly across

a temperature schedule, builds a pairwise similarity matrix over the samples, and aggregates each sample’s similarity row into a self-consistency confidence. The winning sample supplies the label and its aggregated confidence supplies p^* , so the downstream model is unchanged.

3.3 THE FACTUALITY NETWORK AND ITS VARIANTS

The labelled graph becomes a pairwise MRF with one binary variable per claim and per evidence passage. Unary factors are $\phi_X(x) = [1 - \pi_X, \pi_X]$ with π_X the prior on X ; each retained relation contributes one pairwise factor from Table 1 below, oriented so the evidence passage is the premise. Marginals $\mathbb{P}(A_i=1)$ are computed by weighted mini-bucket elimination (Liu & Ihler, 2011) with i-bound 6, and a claim is reported supported when $\mathbb{P}(A_i=1) > \frac{1}{2}$.

Three graph variants differ in which factors are admitted: FR1 connects each claim only to the passages retrieved for it; FR2 deduplicates passages and scores every claim against all of them; FR3 adds evidence–evidence factors. We use FR2. Because scoring every claim against every passage is $O(nm)$ LLM calls, a candidate-pair policy can restrict the pairs actually scored; the calibration of that policy is instructive for the analogous choice in §4.5. Replaying policies against recorded verdicts on a 20-claim narrative with 60 passages, a provenance-restricted policy with an embedding similarity gate holds recall at 1.000 through a threshold of 0.20, slipping to 0.985 at 0.30 and 0.833 at 0.40. Substituting token-Jaccard similarity for sentence embeddings, however, loses 22 of 72 non-neutral relations and moves the factuality score by 0.05 — and *no* threshold recovers full recall, even 0.05. Cheap surrogates for semantic similarity are not safe here, and the saving is workload-dependent rather than constant: roughly $5\times$ on unrelated subtopics but $1.2\times$ on a narrative where every claim shares characters.

Conventions. We use the with-priors factor tables (Table 1) throughout, with $\pi_C = 0.9$ for evidence and $\pi_A = 0.5$ for claims. The published factuality model uses the no-priors tables and a higher evidence prior. The difference is immaterial to the coherence results below, which set claim priors explicitly, but the convention must be stated because the two table families are not interchangeable — that is the subject of §4.3.

4 STAGE TWO: THE COHERENCE MARKOV RANDOM FIELD

4.1 THE MODEL

Definition 1 (Coherence MRF). *Let $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ be the atomic claims of a response in assertion order, each a binary variable with $a_i = 1$ read as “the claim holds”. Let $\mathcal{R} = \{(s_r, t_r, \tau_r, p_r)\}_{r=1}^{|\mathcal{R}|}$ be a set of relations, where $s_r \neq t_r$ index claims, τ_r is an inferential coupling (Def. 2), and $p_r \in [0, 1]$ is the relation’s weight. Given priors $\pi \in [0, 1]^n$, the coherence MRF induced by $(\mathcal{A}, \mathcal{R}, \pi)$ is*

$$\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \prod_{i=1}^n \phi_i(a_i) \prod_{r \in \mathcal{R}} \psi_{\tau_r, p_r}(a_{s_r}, a_{t_r}), \quad Z = \sum_{\mathbf{a} \in \{0,1\}^n} \prod_i \phi_i(a_i) \prod_r \psi_r, \quad (2)$$

where $\phi_i(a_i) = [1 - \pi_i, \pi_i]$ over the states $(0, 1)$ and $\psi_{\tau, p}$ is given by Table 1.

Three structural properties of Definition 1 deserve statement, because each is a deliberate commitment rather than an implementation detail.

Remark 1 (The factor family is closed). *A coupling is nothing more than a choice of which cells of a 2×2 table are down-weighted and by how much. Every $\psi_{\tau, p}$ in Table 1 is determined by p , the source prior π_s , and the penalized set; the whole inferential vocabulary is therefore a three-parameter pairwise family. This is a guarantee about what the model cannot do: no relation type, however elaborately named at the discourse level, can smuggle in expressiveness beyond a pairwise potential. Appendix C contrasts this with Markov logic, whose rule vocabulary is open by design.*

Remark 2 (Order-sensitivity enters only through the relations). *Nothing in Eq. 2 references claim position. Assertion order enters exclusively through which pairs are related and in which direction. It follows that an edit which preserves the relation set — reordering claims whose relations are symmetric, or exchanging a temporal **Precedence** for a **Succession**, which Table 2 maps to no factor at all — provably cannot change any readout. This is a falsifiable prediction about the measure and not merely a design note; §7 describes the ladder built to test it.*

Remark 3 (Coherence is evaluated without evidence). *The graph of Definition 1 contains claim variables only. No retrieved passage appears, and with uniform priors $\pi_i \equiv \frac{1}{2}$ the model consults no external knowledge whatever: it scores how the response hangs together, not whether it is true. This is what makes the two axes separable, and §6 shows how to couple them when a joint reading is wanted.*

4.2 LEVEL ONE: INFERENTIAL COUPLINGS

Definition 2 (Inferential couplings). *A coupling τ is specified by the set $V(\tau) \subseteq \{0, 1\}^2$ of joint assignments (a_s, a_t) it penalizes — equivalently, the configurations at which the relation is violated. The vocabulary is:*

Coupling	Reading	Directed	Penalized set $V(\tau)$
ENTAILMENT	$a_s \Rightarrow a_t$	yes	$\{(1, 0)\}$
CONTRADICTION	$a_s \Rightarrow \neg a_t$	yes	$\{(1, 1)\}$
EQUIVALENCE	$a_s \Leftrightarrow a_t$	no	$\{(0, 1), (1, 0)\}$
EXCLUSIVE	<i>exactly one of a_s, a_t holds</i>	no	$\{(0, 0), (1, 1)\}$
CO_NECESSITY	<i>at least one of a_s, a_t holds</i>	no	$\{(0, 0)\}$
NONE	<i>no coupling</i>	—	$\emptyset$ (no factor)

Write $\mathcal{T}$ for the five edge-producing couplings and $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$ for the conflict couplings, those whose penalized set contains the both-true world $(1, 1)$.

The first three couplings are exactly the NLI edge types of §3, which is what lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose, and the distinction between them and CONTRADICTION is the one the taxonomy exists to draw.

Why exhaustiveness needs its own coupling. CONTRADICTION forbids only the both-true world: two claims that cannot both hold, but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what *exhaustive* alternatives assert. “No one was harmed” and “three people died” can neither both hold nor both fail — either someone was harmed or no one was — so a model that treats them as mere opposition has mis-described the situation in a way that matters for scoring: it leaves available a world in which *neither* claim is true, and so under-penalizes. Conversely CO_NECESSITY is the dual, penalizing only the both-false world, and is what a disjunction asserts (“at least one of these two findings must hold”). Adjudicated disputes — incident investigations, litigation, competing hypotheses — are precisely the domains where exhaustive pairs occur, because the function of a finding is to close one.

Example 1 (A CO_NECESSITY factor in isolation). *Take a single CO_NECESSITY relation with $p = 0.9$ over two claims with priors $\pi = 0.5$, and no other factors. From Table 1 the pairwise table is $[1 - p, \pi_s, \pi_s, p] = [0.1, 0.5, 0.5, 0.9]$, and multiplying in the unaries and normalizing gives world masses*

$$\mathbb{P}(0, 0) = 0.05, \quad \mathbb{P}(0, 1) = 0.25, \quad \mathbb{P}(1, 0) = 0.25, \quad \mathbb{P}(1, 1) = 0.45.$$

The both-false world is suppressed from the 0.25 it would receive under independent fair priors to 0.05, while the three worlds satisfying “at least one holds” share the recovered mass in proportion to how many endpoints they assert. Each marginal rises from 0.5 to 0.7: asserting that at least one of two claims holds is evidence for both, which is the intended reading.

4.3 THE PAIRWISE POTENTIALS, AND WHY FACTUALITY’S TABLES ARE WRONG HERE

Proposition 1 (Row-stochasticity). *Each with-priors table in Table 1 is row-stochastic in the source state: for $\tau \in \{\text{ENTAILMENT}, \text{CONTRADICTION}, \text{CO_NECESSITY}\}$ both rows sum to one, giving the conditional readings*

$$\mathbb{P}(a_t=1 \mid a_s=0) = \pi_s, \quad \mathbb{P}(a_t=1 \mid a_s=1) = \begin{cases} p & \tau = \text{ENTAILMENT} \\ 1 - p & \tau = \text{CONTRADICTION} \\ p & \tau = \text{CO_NECESSITY}. \end{cases}$$

Consequently the factor asserts something about the target only when the source is true; when the source is false, the target is returned to its prior.

Table 1: Pairwise potentials $\psi_{\tau,p}(a_s, a_t)$, listed row-major over the assignments $(0,0), (0,1), (1,0), (1,1)$; π_s is the source claim’s prior and bold marks the penalized set $V(\tau)$ of Def. 2. The *no-priors* family is correct for claim–evidence factuality, where the source passage’s own truth is at stake; the *with-priors* family is what coherence requires (Prop. 1) and what Def. 1 uses. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

Coupling τ	No priors	With priors	Symmetry
ENTAILMENT	$[p, p, \mathbf{1-p}, p]$	$[1-\pi_s, \pi_s, \mathbf{1-p}, p]$	asymmetric
CONTRADICTION	$[p, p, p, \mathbf{1-p}]$	$[1-\pi_s, \pi_s, p, \mathbf{1-p}]$	asymmetric
EQUIVALENCE	$[p, \mathbf{1-p}, \mathbf{1-p}, p]$	$[p, \mathbf{1-p}, \mathbf{1-p}, p]$	symmetric
EXCLUSIVE	$[\mathbf{1-p}, p, p, \mathbf{1-p}]$	$[\mathbf{1-p}, p, p, \mathbf{1-p}]$	symmetric
CO_NECESSITY	$[\mathbf{1-p}, p, p, p]$	$[\mathbf{1-p}, \pi_s, \pi_s, p]$	symmetric

Proposition 1 is the substantive modelling choice of this section, and the contrast with the no-priors family is worth being precise about. In the no-priors ENTAILMENT table the $(1,0)$ cell is $1-p$ and the $(0,\cdot)$ cells are both p , so the table charges the *source* being true whenever the target is uncertain. That is appropriate for claim–evidence factuality: there the source is a retrieved passage whose own reliability is in question, and an entailment from a passage to a claim the model doubts is evidence against the passage. It is wrong for coherence, where the source is another claim of the same response, and where a well-supported premise must not be penalized for being a premise.

The consequence is not a matter of degree. On the response of §9 — sixteen claims joined by a twelve-edge entailment backbone — the no-priors tables drive the root of the backbone to $\mathbb{P}(a_1=1) = 0.19$, far below its 0.5 prior, purely because it is the premise of a chain whose later members are uncertain. The mean marginal sits at 0.490 and moves only to 0.498 when every conflict edge is deleted, a range of 0.008. Under the with-priors tables the same comparison runs 0.607 to 0.620: a range more than 1.5 times larger, from a base that is not already depressed by the backbone itself. All four values are exact, by 2^{16} enumeration. A coherence measure built on the factuality tables does not merely give different numbers; it spends its dynamic range penalizing premises and has little left for conflicts, which are the thing being measured.

4.4 LEVEL TWO: DISCOURSE SENSES AND THE COMPILE MAP

Couplings are the right vocabulary for a factor table and the wrong one for annotation. Asked whether two claims stand in ENTAILMENT, an annotator — human or model — has been handed a question about logic, when the fact actually observable in the text is a question about discourse: does this sentence give a cause, offer evidence, concede a tension, or lay out alternatives? We therefore mine an interpretable layer and compile it down.

Definition 3 (Compile map). *Let $\mathcal{S}$ be the set of Level-2 discourse senses. The compile map*

$$\mathcal{C} : \mathcal{S} \longrightarrow (\mathcal{T} \cup \{\text{NONE}\}) \times ([0, 1] \cup \{\perp\}) \times \{0, 1\}^3$$

assigns to each sense a coupling, a strength prior ($\perp$ meaning “use the estimator’s own value”), and the flags directed, concession and ordering-only. It is total, deterministic, and given in Table 2.

Several entries encode judgements worth defending. **Restatement** carries a fixed strength prior of 0.90 because a marked paraphrase (“in other words”, “equivalently”) is a claim about identity of content, and a model asked to estimate its strength has little to add; the estimator’s value is used when available, and the prior seeds it otherwise. Five senses collapse onto ENTAILMENT, which is a deliberate loss: the model does not distinguish causation from evidence, because a pairwise potential over truth values cannot. What survives compilation is the inferential force, and the sense is retained alongside it for interpretability and for error analysis.

The **Contrast/Alternative** boundary is exactly the exhaustiveness question of §4.2, and the two ordering-only senses are the taxonomy’s way of representing a relation the text genuinely draws while asserting that it has no truth-functional content: a narrative that says one event preceded another has related the two claims, and has said nothing about whether either is true.

Table 2: The compile map $\mathcal{C}$ of Def. 3. *Ordering-only* senses record assertion order but induce **no factor**, which is what makes an order-only edit provably score-invariant (Rem. 2). **Concession** is a conflict the text may itself resolve, handled by Eq. 3. The nine senses marked $\star$ form the admissible vocabulary of §4.5.

Level-2 sense	$\mathcal{C}$ coupling	prior	dir.	flag	gloss
Cause-Effect $\star$	ENTAILMENT	$\perp$	yes		source brings target about
Effect-Cause $\star$	ENTAILMENT	$\perp$	yes		source is the effect, target the cause
Evidence $\star$	ENTAILMENT	$\perp$	yes		source is evidence for target
Condition	ENTAILMENT	$\perp$	yes		source conditions target
Instantiation	ENTAILMENT	$\perp$	yes		target instantiates source
Restatement $\star$	EQUIVALENCE	0.90	no		definitional identity
Contrast $\star$	CONTRADICTION	$\perp$	no		opposed, <i>not</i> exhaustive
Concession $\star$	CONTRADICTION	$\perp$	yes	conc.	a tension the text resolves
Alternative $\star$	EXCLUSIVE	$\perp$	no		exhaustive: exactly one holds
Disjunction $\star$	CO_NECESSITY	$\perp$	no		at least one holds
Precedence $\star$	NONE	—	yes	ord.	temporal order only
Succession	NONE	—	yes	ord.	temporal order only

Concession: a conflict the text resolves. A Concession asserts a tension and then overrides it. “The airline argued pilot error, but the final report found a metallurgical defect” is not an incoherence if the report’s finding is presented as authoritative; treating it as a live contradiction would penalize a response for doing the very thing careful argument requires. When a resolving *holding* claim h is present we discount the conflict’s weight,

$$p_r^{\text{eff}} = p_r (1 - \lambda \pi_h), \quad (3)$$

with $\lambda = 0.45$; absent a resolver the relation stands at full strength and the tension counts against the response. Holdings are identified by adjudicative cue phrases (*held, ruled, concluded, found that, the tribunal, ultimately*) on a claim that is not itself an endpoint of the relation and that maximizes lexical overlap with the two endpoints. This is a heuristic, and with λ it is the one place in the design where hand-set constants enter; §10 returns to the point.

4.5 ESTIMATING RELATIONS FROM THE RESPONSE

Definition 1 takes $\mathcal{R}$ as given. In use it must be estimated from the text, and the estimator is the weakest link in the pipeline (§8), so we specify it carefully.

The estimand, decomposed. A relation’s weight factors into a type term and a strength term:

$$p_r = \underbrace{\mathbb{P}(\tau \mid a_s, a_t, y)}_{\text{type confidence}} \cdot \underbrace{\mathbb{P}(a_t \mid a_s, \tau, y)}_{\text{conditional strength}}. \quad (4)$$

The first factor asks how confident we are that this is a τ -relation *at all*; the second, given that it is, how strongly the source determines the target. Conflating them is a mistake: a confidently-identified weak causal link and a doubtfully-identified strong one are different epistemic situations that a single elicited number cannot distinguish. Two prompts estimate them separately (full texts in Appendix A).

Prompt A elicits the discourse sense and its coupling, with chain-of-thought *preceding* the label so the model commits to a sense only after reasoning — and so that the label’s token logprobs, read exactly as in Eq. 1, measure a decision already made rather than one being made. *Prompt B* elicits the conditional strength as a surrogate token: the answer’s first word must be **Yes** or **No**, and the strength is the renormalized $\mathbb{P}(\text{Yes}) / (\mathbb{P}(\text{Yes}) + \mathbb{P}(\text{No}))$ (Kadavath et al., 2022), or the affirmative fraction over N samples where logprobs are unavailable. The question is deliberately graded — “is the implication at least plausible?” rather than “is it certain?” — so that a weak but real link is not forced to answer **No**; its weakness surfaces in the renormalized probability instead. A verbalized variant asking for $[p=0 . \text{NN}]$ directly is retained only as a comparison, verbalized confidence being weakly calibrated (Xiong et al., 2024).

Grounding is mandatory, not optional. Prompt A always receives the full response, and is instructed to assert a coupling only where the response *itself* draws the connection. This is the single most consequential decision in the estimator. Judging a claim pair in isolation invites the model to accept any abstractly plausible relation, and the failure is not a mild over-generation: measured relation density reaches six to nine relations per claim, including spurious contradictions on paragraphs that contain none. Because Eq. 2 is a product over relations, a graph of many soft mutually-constraining edges drives every marginal toward the prior, so an over-connected graph does not merely add noise — it destroys the measure’s ability to distinguish a coherent response from an incoherent one, in the direction of calling everything mediocre.

Two further precision mechanisms are used, one requested and one checkable. The *admissibility filter* restricts output to the nine (sense, coupling) combinations marked $\star$ in Table 2. Condition and Instantiation are the notable exclusions: both compile to ENTAILMENT and both are semantically broad, so a model reaching for “these two claims are related somehow” lands on them readily — they accounted for 13% of one measured arm’s edges while appearing zero times in gold. The *evidence-span requirement* obliges the model to quote, verbatim from the response, the words that draw the link; the parser then verifies that the quotation is actually a substring of the response and drops the relation otherwise. This is the only precision mechanism in the estimator that is *checked* rather than merely asked for. It must quote the response and not the claims, because claims are decontextualized rewrites and only about 27% of them appear in the response verbatim even after normalization.

Two prompt generations. The first-generation Prompt A over-asserts badly: against a corpus in which 7.2% of candidate pairs are genuinely related, it answers “related” on 72% of the pairs a windowed policy hands it. Five properties push it that way, and the revision addresses each. (i) Its few-shot balance was three related to one unrelated; class balance in the exemplars is a strong prior on the answer distribution, and this one pointed the wrong way by an order of magnitude. The revision is two related to three unrelated. (ii) Its single negative exemplar had the response *explicitly disavow* a link (“the two processes were not related this year”), which teaches that NONE requires denial, whereas real prose simply stays silent; the revision’s negatives are all silence. (iii) It asked for grounding but gave the model nowhere to show it and nothing to check it — hence the evidence span. (iv) Its first reasoning step was a leading question listing eight affirmative relation types before mentioning NONE; the revision asks the discriminating question first, whether the text draws the link or the claims are merely near each other. (v) It gave NONE one line against nine for relation-bearing senses; the revision gives it first position and equal prose. The revision also names *transitivity* explicitly as a non-relation, since roughly 85% of the first generation’s false positives are the model closing a chain the text laid out pairwise, and 96.6% of them touch an endpoint of a real relation. Both generations are retained permanently so that any claim about the prompt can be measured rather than asserted; §8 reports both.

Candidate pairs. Querying all $n(n-1)$ ordered pairs is quadratic in LLM calls and, as argued above, actively harmful. We define four policies:

- ALL_PAIRS: every ordered pair $i \neq j$.
- WINDOWED: forward pairs with $0 < j - i \leq w$, default $w = 4$.
- GATED: the window, plus long-range forward pairs whose embedding or entity similarity exceeds a threshold, to catch discourse callbacks.
- BIDIRECTIONAL: *both* arc directions within $|j - i| \leq w$.

The last exists because of a structural limitation, not a tuning preference. A forward-only policy *cannot emit* an arc that runs backward in claim order, so a gold relation that is both directed and backward is unreachable under it no matter what the model says. In our corpus 20 of 86 edge-producing gold relations are of that kind, capping forward-only recall at 76.7% before the model is consulted; §8 measures the effect.

Every policy is then refined against the response itself (Algorithm 1). Each claim is aligned to its originating sentence; a pair is *discourse-adjacent* if the two claims share content tokens above a Jaccard threshold, if their sentences are within a short span, or if the target’s sentence opens with a discourse connective. Out-of-window pairs that are discourse-adjacent are *promoted* into the

Algorithm 1 SELECTCANDIDATEPAIRS: response-anchored candidate selection.

Require: claims $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ in assertion order; response y ; policy; window w ; gate threshold θ_g ; discourse thresholds θ_d, δ

Ensure: ordered candidate pairs P , plus a coverage record

- 1: **if** policy = ALL_PAIRS **then return** every ordered pair $(a_i, a_j), i \neq j$
- 2: $W \leftarrow \{(a_i, a_j) : 0 < j - i \leq w\}$ $\triangleright$ forward order window
- 3: **if** policy = BIDIRECTIONAL **then** $W \leftarrow W \cup \{(a_j, a_i) : 0 < j - i \leq w\}$
- 4: $G \leftarrow \emptyset$
- 5: **if** policy = GATED **then**
- 6: $G \leftarrow \{(a_i, a_j) : j - i > w, \text{sim}(a_i, a_j) \geq \theta_g\}$
- 7: align each claim to its best-matching sentence of y
- 8: $Prom \leftarrow \{(a_i, a_j) \notin W \cup G : \text{DISCOURSEADJACENT}(i, j)\}$
- 9: $Dem \leftarrow \{(a_i, a_j) \in W : \neg \text{DISCOURSEADJACENT}(i, j)\}$
- 10: **return** $(\text{DEDUP}(W \cup G \cup Prom) \setminus Dem), (|W|, |G|, |Prom|, |Dem|)$

candidate set; in-window pairs that are not are *demoted* out of it. The refinement is a correlated filter — it selects for the same surface adjacency that the sense prompt reads as evidence of relatedness — which is why we report it as a policy component and measure its effect rather than treating it as a neutral preprocessing step.

Failures must be counted, not absorbed. Mining issues one LLM call per candidate pair for Prompt A and one more for each pair that yields an edge, run concurrently under a rate limit. A throttled or failed call returns no parse, and a parser that reads “no parse” as “no relation” converts an infrastructure failure into a sparse graph — which then scores as a *more* coherent response, since fewer conflicts are asserted. We therefore tally call errors separately from genuine NONE answers and report them, and distinguish an explicit NONE from an unparseable coupling for the same reason.

5 FOUR COHERENCE READOUTS

Definition 1 specifies a distribution; a coherence score is a functional of it, and several are defensible. We state five criteria, define four candidates with their exact constructions, and then decide between them by exact inference rather than by preference.

Definition 4 (Criteria). *A coherence readout should be: (R1) a genuine functional of the joint $\mathbb{P}(\mathbf{a})$, not of the relation list alone; (R2) valued in $[0, 1]$ and interpretable; (R3) monotone — resolving or removing a conflict must not decrease it; (R4) sensitive to conflict without hand-tuned constants; (R5) computable from standard inference queries.*

R1 deserves comment, since it is the criterion that does the most work. A readout that depends only on $\mathcal{R}$ — how many relations there are, what types, what weights — is not measuring the response; it is measuring its own annotation. Proposition 2 below shows this is not a hypothetical concern.

5.1 (A) MEAN POSTERIOR MARGINAL

$$\text{LCS}_{\text{mm}}(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(a_i = 1). \quad (5)$$

The marginals come directly from a single MAR query. The interpretation is the expected fraction of the response’s claims that the argument, taken as a whole, upholds. A conflict suppresses the marginals of its endpoints; a satisfied entailment chain propagates support along itself; a claim no relation touches stays at its prior and contributes neutrally.

Alongside the scalar we report three diagnostics that come free with the same query: the per-claim marginals themselves, which localize an incoherence to specific claims; the count of claims dragged *below their own prior*, which is the sharpest single indicator of an active conflict; and the mean normalized binary entropy $\mathcal{E} = \frac{1}{n} \sum_i H_2(q_i)$, lower being more decisive.

5.2 (B) CONSISTENCY: A TWO-TERM PROBABILITY

The obvious alternative to averaging marginals is to ask for the probability that the response is consistent. Making that precise turns out to require two terms:

$$\text{LCS}_{\text{cn}} = \frac{1}{2}(\mathcal{K} + \mathcal{S}), \quad \mathcal{K} = \mathbb{P}\left(\bigwedge_{r: \tau_r = \text{CONTRADICTION}} \neg(a_{s_r} \wedge a_{t_r})\right), \quad \mathcal{S} = \frac{\sum_{r \in \mathcal{R}_S} p_r U_r}{\sum_{r \in \mathcal{R}_S} p_r}, \quad (6)$$

where $\mathcal{R}_S$ ranges over $\{\text{ENTAILMENT}, \text{EQUIVALENCE}, \text{EXCLUSIVE}\}$ and U_r is the probability that relation r is *actively upheld*:

$$U_r = \mathbb{P}(a_{s_r}=1 \wedge a_{t_r}=1) \text{ for ENTAILMENT, EQUIVALENCE,} \quad U_r = \mathbb{P}(a_{s_r} \neq a_{t_r}) \text{ for EXCLUSIVE.}$$

The two coupling sets are deliberately different. The conflict event $\mathcal{K}$ keys on **CONTRADICTION** alone, even though $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$. The reason is that an **EXCLUSIVE** spreads its incoherence over *both* same-value worlds, so reading only its both-true half would describe half the coupling and silently report the other half as satisfactory. **EXCLUSIVE** therefore earns credit in $\mathcal{S}$ instead, where its exactly-one world is simultaneously where the exclusion is honoured and where it is informative. **CO_NECESSITY** enters neither term: its defect is the both-false world, which (a)’s marginals already see directly.

Computing $\mathcal{K}$ exactly. $\mathcal{K}$ is an event probability over a conjunction of $|\{r : \tau_r = \text{CONTRADICTION}\}|$ pairwise events, which a naive encoding would represent as a factor of exponential arity. Instead we augment the network with deterministic auxiliary variables: one u_r per conflict edge with a ternary factor enforcing $u_r = a_{s_r} \wedge a_{t_r}$, then a chained running disjunction $U_k = U_{k-1} \vee u_k$ with ternary factors, and finally $\mathcal{K} = \mathbb{P}(U_{|\cdot|}=0)$. Every auxiliary factor is deterministic and at most ternary, so no $2^{|\mathcal{R}|}$ table is ever built and the augmented network is read with one MAR.

Computing $\mathcal{S}$ exactly. $\mathcal{S}$ is not an event probability but a weighted sum of *pairwise joints* — a linear functional of the joint — so it cannot be read off the same accumulator chain. It needs one deterministic indicator variable per supported edge and its own MAR. Because the indicators are mutually independent given the base network, they may be batched across several queries without approximation, which matters only for inference backends that enumerate.

Why “actively upheld” and not “satisfied”. The definition of U_r credits a relation only in the informative world, not in every world where it is technically satisfied. An entailment $a_s \Rightarrow a_t$ is satisfied vacuously whenever $a_s = 0$, and crediting that is fatal:

Proposition 2 (The vacuous-truth trap). *Let $\mathcal{K}' = \mathbb{P}(\text{every relation in } \mathcal{R} \text{ is satisfied})$, the apparently natural extension of $\mathcal{K}$ to all couplings. Then a response with no mined relations satisfies $\mathcal{K}' = 1$, the maximum attainable value.*

Proof. $\mathcal{K}'$ is the probability of an intersection of $|\mathcal{R}|$ events; for $\mathcal{R} = \emptyset$ the intersection is the whole space. $\square$

The proposition is immediate, but its consequence is not, and it is the reason LCS_{cn} has the form it does. Measured on the family of §9, a disconnected bag of claims scores $\mathcal{K}' = 1.000$ while a response whose causal spine is fully satisfied scores 0.221 — an inversion, not a tie. Nor is the defect repaired by count normalization: the geometric mean over relations gives 0.883 for the base response against 0.882 for the coherent one, non-monotone *and* still 1.000 on the bag; the expected fraction of satisfied relations gives 0.857 against 0.852, likewise inverted; and normalizing against a prior ceiling leaves $[0, 1]$ altogether (8.82 against 5.46) while still running backwards. The common cause is that “are the asserted relations honoured?” is a property of $\mathcal{R}$ rather than of y — a violation of R1 — and it is maximized by asserting nothing. Crediting only the informative world removes the free lunch: a bag of claims upholds nothing and scores $\mathcal{S} = 0$.

The two terms are averaged rather than multiplied so that neither can annihilate the other. A live contradiction should not erase a well-supported spine, and a relation-free response should not inherit a perfect score from an empty conflict set.

5.3 (C) A REIFIED COHERENCE NODE

The most direct reading of “is this response coherent?” as a single probability is to introduce a variable for it. Add a binary R with Bernoulli prior ρ and, for each relation, a ternary *vote* factor

$$h_r(R, a_s, a_t) = \begin{cases} 1 & R = 0 \quad (\text{the branch is flat}) \\ 1 & R = 1, (a_s, a_t) \notin V(\tau_r) \\ 1 - p_r & R = 1, (a_s, a_t) \in V(\tau_r), \end{cases} \quad \text{LCS}_{\text{re}} = \mathbb{P}(R = 1). \quad (7)$$

In the $R = 1$ branch each violated relation charges $1 - p_r$, so R behaves as a noisy-AND over relation satisfaction; in the $R = 0$ branch nothing is charged, so $R = 0$ carries no commitment. With no relations R is decoupled and $\text{LCS}_{\text{re}} = \rho$, correctly signalling that nothing was measured. Note that the violated set is $V(\tau)$ from Def. 2, so ENTAILMENT is treated as satisfied at $a_s = 0$ — vacuously — which is where (c) inherits a weaker form of Prop. 2’s problem. Appendix B works a three-claim subnet through by hand, giving $\mathbb{P}(R=1) = 0.453$.

5.4 (D) NORMALIZED LOG-PARTITION

The partition function measures how much probability mass the constraints leave standing, which is a global coherence signal requiring no marginals at all. It is unnormalized, so we grade it between two references built from the same edge skeleton:

$$\text{LCS}_{\text{lp}} = \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}. \quad (8)$$

For the ceiling, $Z_{\max}$ is the partition function of the same network with every conflict edge removed: deleting constraint factors only adds mass, so $\log Z \leq \log Z_{\max}$ always. For the floor we take the MAP world mass of the base network, $Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$.

Lemma 1 (The MAP world mass is a valid, coherence-graded floor). *$\log Z_{\min} \leq \log Z$ for every network, and $Z_{\min}$ is non-increasing in conflict strength.*

Proof. $Z = \sum_{\mathbf{a}} \text{mass}(\mathbf{a})$ is a sum of non-negative terms, so its single largest term is a lower bound. For the second claim, increasing p_r on a conflict edge multiplies the mass of every world in $V(\tau_r)$ by a factor below one and leaves other worlds unchanged, so the maximum over worlds cannot increase. $\square$

Lemma 1 is not a technicality; it replaces two floors that look more natural and are wrong. Retyping every edge to CONTRADICTION, or saturating conflicts to $p = 1$, both fail to bound $\log Z$ below for the row-stochastic tables of Prop. 1: an isolated conflict removes no mass at all under those tables, whereas a mass-concentrating entailment backbone concentrates a great deal, so a network combining a strong backbone with several soft conflicts can have a base $\log Z$ below either candidate floor. We observed exactly this on real graphs, which put Eq. 8 outside $[0, 1]$ until the MAP floor replaced them. In practice we clamp the ratio to $[0, 1]$ regardless, since approximate inference computes $\log Z$ and the MAP bound with finite i-bound and small tolerance violations are possible.

5.5 DECIDING BETWEEN THE FOUR

Table 3 settles the question empirically and Table 4 summarizes. Readouts (b) and (c) both fail R3, and they fail it at the *same* variant — the one where a conceded tension is resolved — for a shared reason that is worth stating precisely, because it looks like an implementation bug and is not.

Proposition 3 (Belief versus event activity). *Softening a resolved conflict via Eq. 3 increases the probability of its both-true world. For a single conflict edge with uniform priors, discounting p from 0.80 to 0.55 moves $\mathbb{P}(a_s=1, a_t=1)$ from 0.10 to 0.225.*

Proof. Under both conflict couplings the $(1, 1)$ cell of Table 1 is $1 - p$, which increases as p falls. Take a lone CONTRADICTION edge with $\pi_s = \pi_t = \frac{1}{2}$, so the with-priors table is $[\frac{1}{2}, \frac{1}{2}, p, 1 - p]$ and each world carries an equal unary factor of $\frac{1}{4}$. The unnormalized masses are then proportional to the table itself, giving $\mathbb{P}(1, 1) = (1 - p) / (\frac{1}{2} + \frac{1}{2} + p + (1 - p)) = (1 - p) / 2$, which is monotone decreasing in p . Evaluating at $p = 0.80$ and $p = 0.55$ gives 0.100 and 0.225. $\square$

Table 3: All four readouts across a coherence-ordered family of five variants of one response, by exact inference over 2^{16} worlds. The variants run from least to most coherent left to right in the intended ordering. Bold marks a value that moves the *wrong* way relative to its predecessor. Only (a) and (d) are monotone throughout.

Variant	(a) LCS_{mm}	(b) LCS_{cn}	(c) LCS_{re}	$\log Z$	(d) LCS_{lp}
<i>worse</i>	0.586	0.778	0.116	-10.54	0.68
<i>base</i>	0.607	0.781	0.148	-9.64	0.78
<i>concession</i>	0.612	0.772	0.134	-9.64	0.80
<i>fixcasualty</i>	0.613	0.764	0.156	-8.95	0.89
<i>coherent</i>	0.620	0.752	0.181	-8.25	1.00
monotone?	yes	no	no	yes	yes

Table 4: The four readouts against the criteria of Def. 4, with the inference queries each requires.

Readout	Queries	R1	R2	R3	R4	R5	Note
(a) mean marginal	MAR	✓	✓	✓	✓	✓	selected
(b) consistency	$\text{MAR} \times 2$	✓	✓	×	✓	✓	concession dip
(c) reified	MAR	✓	✓	×	×	✓	free p ; dip
(d) norm. $\log Z$	PR, MAP	✓	✓	✓	✓	✓	needs 2 references

The consequence sorts the four readouts into two kinds. A readout that asks *is the conflict active* — (b) counts exactly that event, (c) charges $1 - p_r$ for it — must register a softened conflict as *more* often active, and therefore dips. A readout that asks *how strongly should each claim be believed* — (a) through marginals, (d) through mass — sees less suppression and rises. Since resolving a concession is by construction an increase in coherence, a coherence score must track belief rather than event activity. That is the substantive finding of this section, and it is not specific to our parameterization: any readout defined as the probability of a conflict-free configuration will invert under any discount-based treatment of resolved concessions.

A second failure mode of (b). Table 3 shows (b) failing worse than a single dip, and the reason is instructive about $\mathcal{K}$ and $\mathcal{S}$ jointly. In this family both conflicts are EXCLUSIVE rather than CONTRADICTION, so $\mathcal{K} \equiv 1.000$ throughout by the coupling-set decision above, and the score is carried entirely by $\mathcal{S}$. But removing an exclusion removes *support* credit as well as conflict, so the score runs backwards from 0.778 to 0.752 across the whole ladder. On three-coupling variants of the same response, where the conflicts are CONTRADICTION and $\mathcal{K}$ is live, (b) instead runs $0.576 \rightarrow 0.654 \rightarrow \mathbf{0.582} \rightarrow 0.672 \rightarrow 0.752$: correctly ordered apart from the concession dip. Either way it is non-monotone, and the lesson is that $\mathcal{K}$ and $\mathcal{S}$ should be read separately as diagnostics rather than averaged into a headline.

Selection. We report (a) as the headline. It satisfies all five criteria, needs a single MAR query, and is the only one of the four whose value decomposes into per-claim contributions — which is what makes an incoherence *locatable* rather than merely detectable. We compute and report all four, since (d) is a genuine alternative satisfying the same criteria through a different route (mass rather than marginals), and $\mathcal{K}$, $\mathcal{S}$ carry diagnostic information the mean marginal does not expose.

5.6 ALGORITHM AND COST

Algorithm 2 is dominated by steps 4–9: $|P|$ Prompt-A calls plus one Prompt-B call per surviving edge, which the candidate policy keeps near-linear in n rather than quadratic. Inference is negligible by comparison.

Answering all four readouts requires seven inference calls: the base MAR and PR, the $\mathcal{K}$ accumulator MAR, the $\mathcal{S}$ indicator MAR, the reified MAR, the conflict-free ceiling PR, and the base MAP. Computed independently they would cost fourteen, since each readout re-runs the shared base pair. Production inference is weighted mini-bucket (Liu & Ihler, 2011) with i-bound 6; every *exact* value reported

Algorithm 2 COMPUTELCS(y) — the end-to-end coherence score.

Require: response y ; relation estimator $\mathcal{M}$; discount λ ; priors π
Ensure: LCS $\in [0, 1]$, marginals $\{q_i\}$, diagnostics

- 1: $\mathcal{A} \leftarrow \text{ATOMIZEANDDECONTEXTUALIZE}(y)$
- 2: $P \leftarrow \text{SELECTCANDIDATEPAIRS}(\mathcal{A}, y, \dots)$ ▷ Algorithm 1
- 3: $\mathcal{R} \leftarrow \emptyset$
- 4: **for all** $(a_i, a_j) \in P$ **do** ▷ concurrent, rate-limited
- 5: $(\text{sense}, \tau, c) \leftarrow \text{PROMPTA}(a_i, a_j, y)$ ▷ $c = \mathbb{P}(\tau \mid \cdot)$ from Eq. 1
- 6: **if** $\tau = \text{NONE}$ **then continue**
- 7: $\sigma \leftarrow \text{PROMPTB}(a_i, a_j, \tau, y)$ ▷ surrogate-token strength
- 8: $\mathcal{R} \leftarrow \mathcal{R} \cup \{(i, j, \tau, \text{clip}_{[0,1]}(c \cdot \sigma))\}$
- 9: **for all** $r \in \mathcal{R}$ with $\tau_r \in \mathcal{T}_{\text{conf}}$ sensed as **Concession** **do**
- 10: **if** a resolving holding h exists **then** $p_r \leftarrow p_r(1 - \lambda\pi_h)$
- 11: **else** mark r unresolved
- 12: $\mathcal{N} \leftarrow$ network of Eq. 2 over $(\mathcal{A}, \mathcal{R}, \pi)$ ▷ with-priors tables
- 13: $\{q_i\} \leftarrow \text{INFER}(\mathcal{N}, \text{MAR})$
- 14: **return** $\frac{1}{n} \sum_i q_i, \{q_i\}, (\#\{q_i < \pi_i\}, \mathcal{E}, \log Z)$

in this paper instead comes from explicit 2^n enumeration, which we use as a regression oracle for $n \leq 20$ and which is the source of Tables 3 and the values in §9.

6 COMPOSING COHERENCE WITH FACTUALITY

The priors π of Definition 1 are a free slot, and the natural thing to put in them is what Stage One already computed (Figure 1):

$$\pi_i \longleftarrow q_i = \mathbb{P}(a_i = 1 \mid C). \quad (9)$$

A claim is then credited when it is both externally supported and internally upheld, and the two readings compose in the natural direction: an unsupported claim enters the coherence graph already doubted, so an argument resting on it is penalized twice — once for the evidence and once for what the argument does with it. Setting $q_i \equiv \frac{1}{2}$ recovers the pure coherence score of Remark 3, so the composition is an option rather than a commitment and the two axes remain separately reportable.

Two stages rather than one joint network. One could instead build a single MRF over claim and evidence variables with claim–claim edges added. We do not, for three reasons. Inference runs over n variables rather than $n(1 + k)$ for k retrieved passages per claim, which matters because the augmented networks of §5.2 add further variables on top. The mined relation graph can be rescored under several prior sets — uniform, factuality-derived, ablated — without re-running the estimator, which makes prior ablations essentially free. And the factuality stage stays independently cacheable and independently reportable, which a fused model forfeits. Priors are applied at scoring time and never by mutating the mined graph, so one mined graph supports every prior condition.

Aligning claims across stages. The two stages must agree on which claim is which, and the obvious key is the wrong one. Matching proceeds by object identity first (the shared-atomization path, where the response is decomposed once and both stages see the same objects), then by normalized text (casefolded, whitespace-collapsed, terminal punctuation stripped), and only then by identifier. Text precedes identifier deliberately: near-duplicate removal keeps the first-seen claim and leaves the surviving identifier set *sparse* — $a_0, a_1, a_3, \dots$ — so two independent decompositions of the same response can disagree about which claim occupies which index. Matching on identifiers alone would then attach a prior to the *wrong* claim, which is strictly worse than attaching none, since it silently corrupts the score rather than degrading it. When coverage falls below one half the run is flagged degraded, and the configured policy is to warn, to raise, or to discard every prior and fall back to a clean coherence-only score rather than report a half-primed mixture.

7 EVALUATION PROTOCOL

Validating a coherence measure requires responses that differ in coherence while holding content fixed. Natural corpora do not supply this: two documents that differ in coherence almost always differ in topic, length and claim set as well, so any score difference is unattributable. Human coherence ratings supply a target but not a controlled contrast. And no existing resource supplies what separates *scoring* error from *estimation* error — a gold relation graph, against which the estimator of §4.5 can be measured directly.

We therefore evaluate on synthesized *coherence ladders*. A ladder is a family of five responses over one claim set: a base response realizing a declared relation plan, and four variants produced by applying perturbation operators to that plan — adding a spurious relation, resolving a conceded tension, dropping conflicts — each variant carrying *its own* gold relation graph derived from the base graph by the same operators. Each family declares the ordering its rungs must exhibit, in three classes: **C1**, a strict increase between consecutive rungs; **C2**, a *predicted* decrease or invariance — which is where Prop. 3’s concession dip becomes a falsifiable prediction rather than an excuse, and where an ordering-only edit must be exactly flat by Remark 2; and **C3**, endpoint separation between the extreme rungs. A measure is scored on the fraction of declared constraints it satisfies.

Arms. An *arm* fixes where the relation graph comes from while holding the claims, the priors and the readouts identical. The **GOLD** arm takes each item’s own labelled relations; **GOLD_VALID** keeps only those labelled valid; a **MINED** arm runs the estimator of §4.5 on the response text. This is the central experimental device of the paper. Because the readouts are byte-identical across arms, any shortfall of a mined arm below the gold arm is attributable to relation estimation and not to the probabilistic model, and the two error sources never have to be disentangled after the fact.

Match levels. Mined relations are compared to gold at three nested levels: **pair** (were these two claims related at all), **coupling** (pair, plus the Level-1 coupling, with direction required for the asymmetric couplings), and **sense** (plus the Level-2 sense). The coupling level is the one that matters, since it is the level at which Table 1 selects a factor, and it is materially stricter than pair level.

7.1 PROPOSED BASELINES

A coherence measure should be compared against controls chosen to fail in informative ways. We propose five families and state what each is diagnostic of.

1. *Ladder-blind controls*: claim count, relation count, response length, and number of perturbation operators applied. These cost nothing and they catch the most likely way a coherence measure is secretly wrong — by tracking edit count or graph density rather than coherence. A measure that cannot beat them on C3, or that fails to stay flat where they do, has not earned its complexity.
2. *Local contradiction detection*: pairwise NLI over claim pairs with no global model, scored as one minus the fraction of pairs labelled contradictory, in the spirit of SelfCheckGPT’s NLI variant (Manakul et al., 2023). This isolates precisely what joint inference adds over local detection. The prediction is that it does creditably where the defect is a *present* contradiction and poorly where the defect is a *missing* entailment, since a broken chain has no contradictory pair to find.
3. *LLM judges*: G-Eval’s coherence dimension (Liu et al., 2023) and a direct “rate the logical coherence from 1 to 5” prompt. These are the practical incumbent and the comparison reviewers will want. The ladder is a demanding test for them, since consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while remaining flat where meaning is preserved.
4. *Discourse representations*: DiscoScore (Zhao et al., 2023), with an entity-grid model (Barzilay & Lapata, 2008) as the classical reference. Both are order-sensitive by construction, so ordering ladders are where they should be strongest and conflict ladders — where the perturbation is semantic rather than positional — where they should be weakest.
5. *Internal ablations*: uniform versus factuality priors (Eq. 9); the four readouts of §5; the four candidate-pair policies; the two Prompt-A generations; and the gold-relation ceiling. These separate the model’s design choices from the estimator’s accuracy.

Table 5: Mined-versus-gold agreement over 86 edge-producing gold relations, micro-averaged over ten items. **Fwd/Bwd** split coupling-level recall by whether the gold relation runs forward or backward in claim order. **Non-rel** counts violations of the 48 pairs the corpus explicitly declares unrelated — a real negative set, unlike the unlabelled remainder. **Illegal** counts edges outside the nine admissible (sense, coupling) combinations, which cannot be scored against gold at all. **Calls** is LLM calls issued.

Policy / prompt	Edges	Pair R	Coup P	Coup R	Coup F1	Fwd	Bwd	Non-rel	Illegal
WINDOWED	190	83.7	25.8	57.0	35.5	75.0	23.3	7/48	25
GATED	321	83.7	14.6	54.7	23.1	69.6	26.7	8/48	38
ALL_PAIRS	867	95.3	9.3	82.6	16.8	87.5	73.3	10/48	93
BIDIRECTIONAL	205	83.7	39.0	66.3	49.1	71.4	56.7	5/48	13
BIDIR., prompt v2	173	73.3	48.3	67.4	56.3	85.7	33.3	0/48	0

Beyond ladder ordering we recommend Spearman correlation against human coherence ratings on the same items, reported alongside per-family ordering agreement. Ordering agreement is the sharper instrument, being a controlled contrast with a known answer; human correlation is what establishes that the construct being ordered is the one readers care about. Neither substitutes for the other.

8 EXPERIMENTS

Scale and sampling, stated first. Two bodies of data are used. The first is nine hand-authored fixtures — a recall report, two biographies (one with five planted contradictions), a narrative, a summary, two summaries of one appellate judgment differing only in claim order, an incident post-mortem, and a damages opinion — carrying claims and prose notes but no machine-readable gold relations. The second is a generated corpus of coherence ladders: **two families, ten items**, produced by `claude-opus-5` with a three-model validation committee (`gpt-5.6`, `gemini-3.1-pro`, `claude-sonnet-5`), on the canonical topics Anthropology and Archaeology, sixteen claims per item. Relations are mined by `llama-3.3-70b-instruct`. Mining is sampled once per cell at the model’s default temperature, and repeated runs of an identical configuration moved coupling-level F1 by up to 0.06; across five independent samples the two revised mining strategies span 0.49–0.58, every one of them above the unrevised 0.36. So the gap to the original exceeds the sampling noise, while the ordering *between* the two revisions does not. These are measurements, not significance claims, and the second decimal place of any single cell should not be read. All exact values are by 2^{16} enumeration; all corpus-scale values are by weighted mini-bucket with i-bound 6.

8.1 RELATION ESTIMATION IS THE BOTTLENECK

Table 5 measures the estimator of §4.5 against gold, and three findings stand out.

First, `ALL_PAIRS` achieves by far the best recall (82.6% at coupling level, 95.3% at pair level) and by far the worst precision (9.3%): querying every pair finds the true relations and buries them, emitting 867 edges against 86 gold. This is not a benign precision/recall trade. Since Eq. 2 is a product over relations, 781 spurious factors do not merely dilute the signal — they drive every marginal toward the prior, which is exactly the collapse §4.5 predicted. The corresponding mean marginal (Table 6) is 0.590, the lowest of any arm including gold, on the same responses.

Second, the forward-only policies are *structurally* capped, as predicted: their backward recall is 23–27% against 73.3% for `ALL_PAIRS`, because 20 of the 86 gold relations are both directed and backward and no forward-only candidate set can emit them. Admitting both arc directions within the window recovers most of that gap (56.7% backward) at 205 edges and 505 calls, against `ALL_PAIRS`’ 867 edges and 3,267 calls — a sixth of the calls for two-thirds the coupling recall and four times the precision.

Third, the prompt revision of §4.5 is where the largest gain lies. Rebalancing the few-shot label distribution and requiring a verifiable evidence span raises coupling F1 from 35.5 to 56.3, drives declared-non-relation violations from 7/48 to 0/48, and drives illegal-vocabulary emissions from 25 to 0 — while *reducing* the number of calls. Precision, not recall, is where a grounded relation estimator is won, and it is won in the prompt rather than in the candidate policy.

Table 6: Left: declared ordering constraints satisfied, 28 assertions per arm (14 per family $\times$ two families). Right: each readout averaged over items. GOLD_VALID is near-flat by construction — a conflict-fixing rung removes the planted-invalid conflicts that this arm has already discarded — so its failures carry no information about the readouts; the GOLD arm is what the constraints test. Averaging a readout across the rungs of a ladder deliberately collapses the ordering the ladder exists to test, so the right-hand block is a level comparison only.

Arm	Constraints satisfied			Mean readout value			
	f001	f002	%	LCS _{mm}	LCS _{cn}	LCS _{re}	LCS _{lp}
GOLD	11/14	11/14	78.6	0.796	0.092	0.232	0.383
GOLD_VALID	7/14	4/14	39.3	0.793	0.157	0.292	0.462
MINED, WINDOWED	4/14	9/14	46.4	0.843	0.007	0.360	0.164
MINED, GATED	7/14	8/14	53.6	0.679	0.002	0.262	0.102
MINED, ALL_PAIRS	7/14	5/14	42.9	0.590	0.163	0.037	0.018
MINED, BIDIRECTIONAL	5/14	5/14	35.7	0.810	0.000	0.403	0.143
MINED, BIDIR. prompt v2	6/14	8/14	50.0	0.660	0.053	0.366	0.088

Table 7: LCS_{mm} on the nine hand-authored fixtures, across three relation-mining models and three conditional-strength estimators (*lp* = surrogate logprobs, *smp* = surrogate sampled, *vr* = verbalized). *e2-bio* is a consistent biography and *e2-con* the same biography with five planted contradictions; *e5-K* and *e5-S* are the two claim-identical orderings of one judgment; *ex6* is an authored contradiction-free post-mortem.

Model	Str.	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6
llama-4-mav.	lp	0.76	0.66	0.88	0.49	0.49	0.51	0.41	0.46	0.95
	smp	0.81	0.72	0.91	0.54	0.57	0.53	0.54	0.33	0.94
	vr	0.75	0.75	0.88	0.53	0.75	0.71	0.90	0.70	0.94
llama-3.3-70b	lp	0.79	0.82	0.99	0.82	0.72	0.32	0.71	0.49	0.93
	smp	0.70	0.89	1.00	0.13	0.69	0.26	0.65	0.57	0.93
	vr	0.65	0.67	0.94	0.46	0.69	0.59	0.67	0.56	0.94
gpt-oss-120b	lp	0.76	0.82	0.89	0.49	0.75	0.79	0.95	0.90	0.95
	smp	0.73	0.93	0.88	0.54	0.74	0.89	0.96	0.89	0.94
	vr	0.71	0.89	0.66	0.45	0.71	0.70	0.86	0.73	0.93

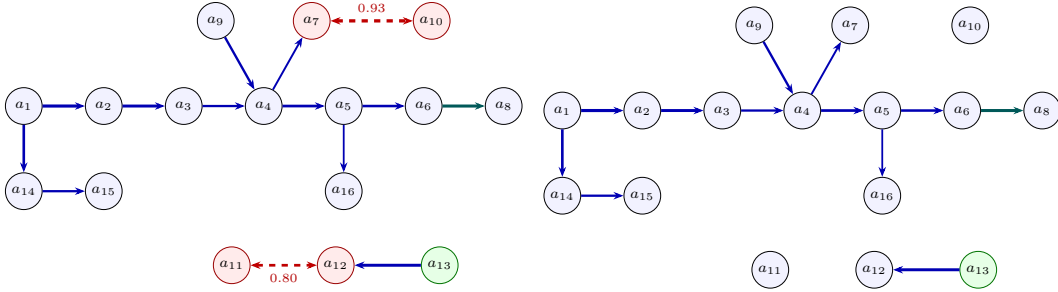
8.2 THE MODEL ORDERS CORRECTLY WHEN THE GRAPH IS CORRECT

On gold relations the readouts satisfy 78.6% of the declared ordering constraints — 22 of 28 (Table 6). Mined arms reach 35.7–53.6%. Since the readouts, the claims and the priors are byte-identical across arms, the 25–43 point gap is attributable to relation estimation, and it is consistent with Table 5: at 56.3% coupling F1 roughly half the factors in a mined network are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper’s central empirical claim, and it is a claim about *where the remaining problem is*: the probabilistic model orders responses correctly when handed a correct graph, and the open problem is extracting one.

Two secondary observations. The ordering of mined arms does not track their coupling F1 — GATED scores best among them at the second-worst F1 — which at 28 assertions per arm we read as sampling noise rather than signal, and which is precisely why policies should not be ranked on ladder agreement at this corpus size. And the mean-readout block shows the ALL_PAIRS collapse quantitatively: LCS_{mm} = 0.590 against gold’s 0.796 on the same ten responses, from adding 781 spurious factors.

8.3 READOUT DYNAMIC RANGE IS A PROPERTY OF THE GRAPH

Table 7 scores the nine fixtures across nine configurations. The consistent biography separates from its contradicted twin in every configuration but one (0.99 against 0.82 for the best-calibrated setting), and the authored contradiction-free post-mortem scores highest in all nine (0.93–0.95), which is



(a) **Incoherent** (base). Two EXCLUSIVE conflicts (dashed): $a_7 \leftrightarrow a_{10}$ is *unresolved*; $a_{11} \leftrightarrow a_{12}$ is a concession the holding a_{13} (green) resolves. $\text{mm} = 0.607$, $\log Z = -9.64$. (b) **Coherent**. The identical support structure with both conflict edges removed; a_{10} and a_7 are no longer suppressed. $\text{mm} = 0.620$, $\log Z = -8.25$, $\text{lp} = 1.00$.

Figure 2: Relation graphs for the AeroParts recall report (§9). Solid blue: ENTAILMENT; teal: EQUIVALENCE; dashed red, double-headed: EXCLUSIVE. Line width grows with the relation weight p . The two panels differ only in the two conflict edges, so the $0.607 \rightarrow 0.620$ lift and the 1.39 nats of recovered mass are attributable to conflict structure alone.

the sanity check the fixture was written for. Aggregated over all cells, contradiction-heavy fixtures average $\text{LCS}_{\text{mm}} = 0.560$ against 0.767 for the conflict-free ones.

The four readouts differ sharply in dynamic range, and the pattern is instructive. Under the response-grounded WINDOWED policy the estimator asserts almost no CONTRADICTION edges on these fixtures, so $\text{LCS}_{\text{cn}} \equiv 1.00$ and $\text{LCS}_{\text{re}} \equiv 0.50$ (spread 0.00) while LCS_{mm} (spread 0.87) and LCS_{lp} (spread 1.00) discriminate. Under ALL_PAIRS the same readouts regain range, LCS_{re} spanning 0.00–0.50. The degeneracy is therefore a property of the *graph*, not of the readout: $\mathcal{K} = 1.00$ is the *correct* answer for a conflict-free response — verified on the post-mortem, where $\text{LCS}_{\text{mm}} = 0.939$ and $\mathcal{K} = 1.000$ at 1.9 relations per claim — and becomes uninformative exactly when the estimator asserts no conflicts at all. This is the practical case for reporting LCS_{mm} as the headline with $\mathcal{K}$ and $\mathcal{S}$ as diagnostics: a number that reads 1.00 both for a sound argument and for an unmined one cannot be used alone.

Finally, the three conditional-strength estimators differ less than the models do: averaged over all model/fixture cells, surrogate-logprobs gives 0.722, surrogate-sampled 0.712, and verbalized 0.730. The verbalized baseline is not obviously worse in *level*, but it is the least stable across fixtures (compare its e5–K column, 0.90/0.67/0.86, against surrogate-logprobs’ 0.41/0.71/0.95), consistent with the calibration literature (Xiong et al., 2024).

9 TWO WORKED EXAMPLES

The response is a sixteen-claim aviation recall report. Its relation graph is twelve support edges — an entailment spine $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_6$ (shipment certified to spec, installation, entry into service, in-flight failures, regulatory probe, grounding order), branches to the pulled-from-service claim, the maintenance-log evidence, the tribunal’s damages award and the regulator’s bulletin, and one EQUIVALENCE edge — plus two conflicts. Every value below is exact, by 2^{16} enumeration, and the graph is reproduced edge-by-edge in Figure 2 with each edge’s weight shown by line thickness.

Example 2 (Incoherent). *The report asserts both that no passengers or crew were harmed and that a wire report claimed three passengers died. These are exhaustive alternatives rather than mere opposition: they cannot both hold and cannot both fail, so the relation is EXCLUSIVE with $p = 0.93$ — and nothing in the response resolves it. A second tension, between the airline’s pilot-error argument and the final report’s metallurgical-defect finding ($p = 0.80$), is a **Concession** that the tribunal’s holding does resolve, and Eq. 3 discounts it accordingly. The model returns $\text{LCS}_{\text{mm}} = 0.607$ with $\log Z = -9.64$. The diagnostic that localizes the incoherence is per-claim: the marginal of the claim on the losing side of the unresolved conflict is dragged below its own prior, which no aggregate score would reveal. Applying the concession discount alone — resolving the second tension while leaving the first — raises the score to 0.612, as C2 requires.*

Example 3 (Coherent). *Removing the two conflict edges and changing nothing else leaves the identical support structure (Figure 2b). The score rises to $\text{LCS}_{\text{mm}} = 0.620$ with $\log Z = -8.25$ and $\text{LCS}_{\text{lp}} = 1.00$: the response is now as coherent as its own edge skeleton permits, which is what the ceiling of Eq. 8 means. The suppressed claim recovers to its prior.*

Because the two panels of Figure 2 differ *only* in the two conflict edges, the $0.607 \rightarrow 0.620$ lift and the 1.39 nats of recovered probability mass are attributable to conflict structure alone — which is the property the ladder construction of §7 generalizes from one pair to a family. The full ladder over this family is monotone throughout: $0.586 < 0.607 < 0.612 < 0.613 < 0.620$ over the rungs *worse, base, concession-resolved, one-conflict-fixed, coherent*.¹

A second fixture pair makes the order-sensitivity claim concretely. Two summaries of one appellate judgment are built from the *same* facts, differing only in whether self-serving assertions precede or follow the findings that qualify them. The faithful ordering scores $\text{LCS}_{\text{mm}} = 0.95$ and the self-serving-first ordering 0.90. No per-claim factuality check can separate them, because every claim is identical; what differs is which claims the response leaves standing unqualified at the point where it asserts them.

10 LIMITATIONS

The model is pairwise, and some incoherence is not. Definition 1 admits only pairwise factors, so genuinely k -ary structure lies outside the family: three claims can be jointly inconsistent with no two of them pairwise inconsistent, and our model cannot represent that. Two specific consequences are worth naming. Transitivity is not enforced — the model will happily hold $a \Rightarrow b$ and $b \Rightarrow c$ at high weight while leaving a and c unrelated, since it has no rule tying them. And a claim conceded by two independent holdings is discounted twice rather than once. Both are expressible as first-order rules in the Markov logic view of Appendix C, which is the natural route past this limitation and which we have not taken.

Hand-set constants. The design deliberately avoids free parameters — criterion R4 exists for that reason, and it is what disqualifies readout (c) with its prior ρ . Two constants remain. The concession discount $\lambda = 0.45$ in Eq. 3 sets how much a resolved tension is softened, and the resolver-detection heuristic identifies holdings by cue phrases and lexical overlap. Both are defensible and neither is derived. A principled treatment would learn the discount from labelled resolved and unresolved concessions, which the corpus of §7 supplies in principle but not at the scale required.

Relation estimation dominates the error. §8 is unambiguous that the bottleneck is not the model. At 56.3% coupling-level F1, roughly half the factors in a mined network are wrong or missing, and the 25–43 point ladder-agreement gap between mined and gold arms follows from that. Everything the paper establishes about the readouts is established on gold or exact graphs; the mined-arm numbers should be read as a measurement of current extraction quality, not as the measure’s ceiling.

Approximate inference is not characterized. Production inference is weighted mini-bucket with i -bound 6, while every exact value reported here comes from 2^n enumeration, feasible only for $n \leq 20$ claims. The two agree on the fixtures we can check both ways, but we have not characterized the approximation error at length, and the augmented networks of §5.2 — which add one auxiliary variable per relation — are the case where it would most plausibly bite.

Empirical scope. The generated corpus is two families and ten items, of one ladder type, with single-sample mining and measured run-to-run movement of 0.06 F1. Three further ladder types are specified — including the control ladder whose whole purpose is to catch a measure that tracks edit count, arguably the most important single experiment in the protocol — but were not generated at scale. The baselines of §7.1 are argued for rather than measured. And the corpus is LLM-generated

¹Treating both tensions as plain CONTRADICTION rather than EXCLUSIVE — the three-coupling model that omits the exhaustiveness distinction — gives $\text{LCS}_{\text{mm}} = 0.587$ and $\log Z = -9.75$ for Example 2, with Example 3 unchanged at 0.620 and -8.25 . The distinction of §4.2 is worth 0.020 of score here, and the reason it matters more than that figure suggests is Table 3: with EXCLUSIVE conflicts, readout (b)’s conflict term is pinned at 1.000 and the readout fails outright.

throughout, gold relation labels included, with LLM-committee validation; a bias shared between generator and committee would not be caught by it.

11 CONCLUSION

Coherence is a property of a joint distribution over what a response asserts, and it can be read off a Markov random field whose factors are the relations the response itself draws. Three things follow from taking that view seriously. First, the factor tables matter: the ones that are right for grounding a claim against evidence charge a premise for being a premise, and a coherence measure built on them spends its dynamic range in the wrong place (Prop. 1). Second, the relation vocabulary can be rich at the annotation layer and must be narrow at the inferential one; twelve discourse senses compiling deterministically onto five couplings, each fixed by the world it penalizes, keeps the model a closed pairwise family while leaving the labels interpretable. Third, and least expected, not every reading of “how coherent is this?” is admissible. Of four candidates, two are non-monotone in conflict resolution for a single diagnosable reason — they measure whether a conflict is active rather than what should be believed (Prop. 3) — and a fifth, apparently the most natural of all, is maximized by asserting no relations whatever (Prop. 2). The mean posterior marginal survives all five criteria, needs one inference query, and localizes an incoherence to the claims responsible.

Measuring such a score requires controlled contrasts, and ladders of responses over one claim set with declared orderings supply them. Given gold relation graphs the readouts recover 78.6% of those orderings; given graphs mined from the response text, 53.6%. Because the two conditions differ in nothing but the relation graph, the gap localizes the open problem: the probabilistic model is adequate, and grounded relation extraction, at 56.3% coupling-level F1, is the bottleneck. That is a more tractable target than “measure coherence”, and it is one that improvements can be measured against.

AI USE STATEMENT

Generative AI is both an object of study and a component of the artifacts released here, and we disclose both roles. The relation estimator of §4.5 is LLM-based by design: its prompts, its logprob-derived confidences and its failure modes are contributions of the paper. The evaluation corpus of §7 is LLM-generated throughout, including its gold relation labels, and validated by an independent LLM committee; §10 states the consequences for treating it as ground truth. We additionally used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that pins the exact numerical values reported in §9 and Table 3; every number in §8 was traced to a stored experiment record. The research questions, the taxonomy, the readout definitions, the propositions and their proofs are the authors’ own. We take responsibility for the final content of this work.

ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. The fixtures include a summary of a published appellate judgment, used as a worked example of ordering-dependent coherence; it is a matter of public record and no private individual is the subject of analysis. The principal ethical risk is misreading: a coherence score is not a truth score, and a response can be perfectly coherent and entirely false. That is precisely why §6 keeps the two axes separable, and we caution against deploying LCS alone as a factuality filter — a fluent, internally consistent fabrication is exactly the artifact it scores highest. The released fixtures and corpus deliberately contain false claims, labelled as such, and should not be redistributed without those labels.

REPRODUCIBILITY STATEMENT

The model is fully specified in the paper: Definition 1 gives the joint, Table 1 both families of potentials, Def. 2 the penalized sets, Table 2 the compile map, and Eqs. 5–8 the four readouts together with the auxiliary-variable constructions two of them require. Algorithms 1 and 2 give candidate selection and the end-to-end procedure. Appendix A reproduces all four estimation prompts verbatim, including the trailing token boundary that the surrogate readout depends on. The exact values in

§9 and Table 3 are computed by 2^{16} enumeration and are pinned by regression tests in the released code, so they are reproducible without LLM access; the relation graph they use is given edge-by-edge in Figure 2. §8 names every model used and states the sampling regime and its measured variance. Code, fixtures and corpus are released.

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A THE ESTIMATION PROMPTS, VERBATIM

The four prompts below are reproduced exactly as they are issued, extracted programmatically from the released source so that this appendix cannot drift from the code. Placeholders in double braces are substituted at call time; `{{sense_menu}}` is substituted at build time from one of the two menus in §A.5.

A.1 PROMPT A, FIRST GENERATION — DISCOURSE SENSE AND COUPLING

6,148 characters. Placeholders: `{{sense_menu}}`, `{{response}}`, `{{atom_a}}`, `{{atom_b}}`.

```

1
2
3 Instructions:
4 You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
5   ↳ taken FROM THAT RESPONSE, in their order of appearance (A comes before B). Your task is
6   ↳ to decide the discourse/logical relation FROM A TO B, following the steps below.
7
8 IMPORTANT -- ground your decision in the response. Assert a coupling ONLY if the response
9   ↳ ITSELF draws that connection between A and B (as written, or as a clear step in the
10  ↳ author's argument/narrative). Do NOT assert a relation that is merely plausible in
11  ↳ general but that the response does not actually make. If A and B both appear in the
12  ↳ response yet the response draws no logical or discourse dependence between them, the
13  ↳ answer is None.
14
15 1. Reason step by step, referring to the response: does the response present A as causing,
16  ↳ enabling, providing evidence for, restating, elaborating, temporally preceding,
17  ↳ contrasting with, or contradicting B? Is B a claim the response later withdraws or that
18  ↳ a holding resolves? Consider the direction (A to B). If the response links A and B only
19  ↳ indirectly through other claims, or not at all, that is None.
20
21 2. Name the DISCOURSE SENSE, one of: {{sense_menu}}.
22   ↳ Cause-Effect: A causes/leads to B. Effect-Cause: A is the effect, B its cause.
23   ↳ Evidence: A provides evidence for B. Condition: A is a condition for B.
24   ↳ Restatement: A and B assert the same thing. Instantiation: A is a general claim, B a
25   ↳ specific instance (or vice versa).
26   ↳ Contrast: A and B are in opposition but NOT exhaustive (they need not cover all
27   ↳ possibilities; both could conceivably be false).
28   ↳ Concession: A and B are in tension but the text concedes/resolves it ("although A,
29   ↳ still B", or a holding settles it).
30   ↳ Alternative: A and B are EXHAUSTIVE competing options -- EXACTLY ONE holds (they are
31   ↳ mutually exclusive AND together cover the possibilities: not both, and not neither).
32   ↳ E.g. "no one was harmed" vs "three people died"; "the cause was pilot error" vs "the
33   ↳ cause was a metallurgical defect".
34   ↳ Disjunction: AT LEAST ONE of A and B holds (they may both hold, but the response rules
35   ↳ out neither being true) -- e.g. two supporting findings at least one of which must
36   ↳ be present.
37   ↳ Precedence/Succession: A and B are ordered in time with no truth dependence.
38   ↳ None: the response draws no logical or discourse dependence between A and B.
39
40 3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive,
41  ↳ co_necessity, none.
42   ↳ Cause-Effect, Effect-Cause, Evidence, Condition, Instantiation -> entailment
43   ↳ Restatement -> equivalence
44   ↳ Contrast, Concession -> contradiction
45   ↳ Alternative -> exclusive (exactly one of A, B is true)
46   ↳ Disjunction -> co_necessity (at least one of A, B is true)
47   ↳ Precedence, Succession, None -> none
48   Prefer "exclusive" over "contradiction" when the two claims are not just incompatible
49   ↳ but EXHAUSTIVE (one of them must be true); prefer "contradiction" when they merely
50   ↳ cannot both hold but could both be false.
51
52 4. Give your final answer as two bracketed tags on ONE line, sense first:
53 [sense=Cause-Effect] [coupling=entailment]

```

```

1188 32 A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.
1189 33
1190 34 Use the following examples to better understand your task.
1191 35
1192 36 Example 1 (the response makes the causal link):
1193 37 RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns
1194 38 ↳ spiked, and the company's stock price fell 15 percent over the same period.
1195 39 A: The company launched a flawed product last quarter.
1196 40 B: The company's stock price fell 15 percent last quarter.
1197 41 1. Reasoning: the response presents the flawed launch as the head of a chain (panning,
1198 42 ↳ returns) that ends in the stock decline, so the response itself draws a causal link
1199 43 ↳ from A to B.
1200 44 2. Discourse sense: Cause-Effect.
1201 45 3. Coupling: A causing B is a positive inferential link, i.e. entailment.
1202 46 4. Final answer:
1203 47 [sense=Cause-Effect] [coupling=entailment]
1204 48
1205 49 Example 2 (both claims present, but the response draws NO connection -> None):
1206 50 RESPONSE: The quarterly report was published in April. Separately, the annual audit was
1207 51 ↳ scheduled for December. The two processes are run by different teams and were not
1208 52 ↳ related this year.
1209 53 A: The quarterly report was published in April.
1210 54 B: The annual audit was scheduled for December.
1211 55 1. Reasoning: both claims appear in the response, and one might imagine a
1212 56 ↳ reporting-to-audit link in general, but this response explicitly treats them as
1213 57 ↳ separate and unrelated. The response draws no dependence from A to B.
1214 58 2. Discourse sense: None.
1215 59 3. Coupling: no dependence the response asserts, i.e. none.
1216 60 4. Final answer:
1217 61 [sense=None] [coupling=none]
1218 62
1219 63 Example 3 (the response states an EXHAUSTIVE alternative -> exclusive):
1220 64 RESPONSE: The official statement said no one was harmed in the incident. However, the
1221 65 ↳ coroner's report confirmed that three people died in the incident.
1222 66 A: No one was harmed in the incident.
1223 67 B: Three people died in the incident.
1224 68 1. Reasoning: the response sets A and B against each other ("However, ...") and they cannot
1225 69 ↳ both be true; but they also cannot both be false -- either people were harmed or they
1226 70 ↳ were not -- so exactly one holds. This is exhaustive, not a mere contrast. No holding
1227 71 ↳ resolves it.
1228 72 2. Discourse sense: Alternative.
1229 73 3. Coupling: exactly one of A, B is true, i.e. exclusive.
1230 74 4. Final answer:
1231 75 [sense=Alternative] [coupling=exclusive]
1232 76
1233 77 Example 4 (at least one must hold -> co_necessity):
1234 78 RESPONSE: The defect was caught in review: at least one of the two independent checks --
1235 79 ↳ the vibration analysis or the metallurgical assay -- flagged it.
1236 80 A: The vibration analysis flagged the defect.
1237 81 B: The metallurgical assay flagged the defect.
1238 82 1. Reasoning: the response asserts the defect WAS caught by at least one check, so A and B
1239 83 ↳ cannot both be false; but both could hold (both checks may have flagged it). This is a
1240 84 ↳ disjunction, not an exclusion.
1241 85 2. Discourse sense: Disjunction.
1242 86 3. Coupling: at least one of A, B is true, i.e. co_necessity.
1243 87 4. Final answer:
1244 88 [sense=Disjunction] [coupling=co_necessity]
1245 89
1246 90 Your task:
1247 91 RESPONSE: {{response}}
1248 92 A: {{atom_a}}
1249 93 B: {{atom_b}}

```

A.2 PROMPT A, SECOND GENERATION — REBALANCED FOR PRECISION

6,361 characters. Same placeholders. The five changes relative to the first generation, and their motivation, are given in §4.5: few-shot balance from 3:1 related to 2:3; negative exemplars showing silence rather than explicit disavowal; a verifiable evidence span; the discriminating question asked first; and NONE given first position and equal prose. Transitivity is named as a non-relation.

```

1238 1
1239 2
1240 3 Instructions:
1241 4 You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
1242 5 ↳ taken FROM THAT RESPONSE. Decide the discourse/logical relation FROM A TO B.

```

Most pairs of claims in a text are NOT related. They are simply near each other. Your default answer is None, and you should depart from it only when the response itself draws a dependence between A and B.

Three things that are NOT relations, however plausible they look:

- Topical adjacency. A and B being about the same subject, or sitting in neighbouring sentences, is not a relation.
- World knowledge. If the link is true in general but the response does not draw it, the answer is None.
- Transitivity. If the response links A to some third claim C, and C to B, that does NOT make A related to B. Report only the links the text draws directly.

1. First ask the discriminating question: does the response ITSELF assert a dependence from A to B -- as written, or as an explicit step in the author's argument? Identify the words in the response that do it. If you cannot point to such words, the answer is None.
2. If and only if you can, name the DISCOURSE SENSE, one of: {{sense_menu}}.
 - None: the response draws no dependence between A and B. This is the common case: unrelated, merely adjacent, related only by world knowledge, or related only through a third claim.
 - Cause-Effect: the response says A causes/leads to B. Effect-Cause: A is the effect, B its cause.
 - Evidence: the response offers A as evidence/support for B.
 - Restatement: A and B say the same thing -- a paraphrase, a converse, or a restatement the text marks ("in other words", "equivalently", "that is").
 - Contrast: A and B are opposed but NOT exhaustive (both could be false).
 - Concession: A and B are in tension and the text concedes or resolves it ("although A, still B"; a holding settles it).
 - Alternative: A and B are EXHAUSTIVE competing options -- exactly one holds (not both, not neither).
 - Disjunction: AT LEAST ONE of A and B holds; both may.
 - Precedence: A and B are ordered in time with no truth dependence.
3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive, co_necessity, none.
 - Cause-Effect, Effect-Cause, Evidence -> entailment
 - Restatement -> equivalence
 - Contrast, Concession -> contradiction
 - Alternative -> exclusive (exactly one of A, B is true)
 - Disjunction -> co_necessity (at least one of A, B is true)
 - Precedence, None -> none
 Prefer "exclusive" when the two claims are incompatible AND exhaustive; "contradiction" when they merely cannot both hold but could both be false.
4. Answer on ONE line, with three bracketed tags. The evidence must be a SHORT VERBATIM QUOTE COPIED FROM THE RESPONSE -- the words that draw the link. Do not quote claim A or claim B; quote the response. Use [evidence=none] when the sense is None. [sense=Evidence] [coupling=entailment] [evidence="which is why the panel held"]

Use the following examples to better understand your task.

Example 1 (adjacent and on-topic, but the response draws no link -> None):

RESPONSE: The survey was administered in March to a sample of 400 households. Response rates in rural districts have declined over the past decade. The analysis weighted results by district population.

A: The survey was administered in March to a sample of 400 households.
 B: Response rates in rural districts have declined over the past decade.

1. The two sentences are adjacent and both concern the survey, but the response never says the March administration bears on rural response rates. There are no words that draw a dependence.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Example 2 (the response makes the causal link explicitly):

RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns spiked, and as a direct result the company's stock price fell 15 percent over the same period.

A: The company launched a flawed product last quarter.
 B: The company's stock price fell 15 percent last quarter.

1. The response presents the launch as the head of a chain ending in the stock decline, and marks it: "as a direct result".
2. Discourse sense: Cause-Effect.
3. Coupling: entailment.
4. [sense=Cause-Effect] [coupling=entailment] [evidence="as a direct result"]

Example 3 (true in general, but this response does not draw it -> None):

RESPONSE: Higher interest rates raise the cost of corporate borrowing. The firm opened three distribution centres last year. Its logistics costs fell by eight percent.

A: Higher interest rates raise the cost of corporate borrowing.
 B: The firm opened three distribution centres last year.

1296 62 1. One could argue borrowing costs affect expansion decisions, but that is world knowledge.
 1297 63 ↳ This response states the two facts side by side and links neither to the other.
 1298 64 2. Discourse sense: None.
 1299 65 3. Coupling: none.
 1300 66 4. [sense=None] [coupling=none] [evidence=none]
 1301 67 Example 4 (the response states an exhaustive alternative -> exclusive):
 1302 68 RESPONSE: The two readings are mutually exclusive and exactly one must be right: either no
 ↳ one was harmed in the incident, or three people died in it.
 1303 69 A: No one was harmed in the incident.
 1304 70 B: Three people died in the incident.
 1305 71 1. The response sets A and B against each other and says exactly one must be right:
 ↳ "mutually exclusive and exactly one must be right: either ... or".
 1306 72 2. Discourse sense: Alternative.
 1307 73 3. Coupling: exclusive.
 74 4. [sense=Alternative] [coupling=exclusive] [evidence="mutually exclusive and exactly one
 ↳ must be right"]
 1308 75
 1309 76 Example 5 (linked only through a third claim -> None):
 1310 77 RESPONSE: The alloy is chemically identical to the certified reference. It therefore meets
 ↳ the reference specification. Meeting that specification is a precondition for airframe
 ↳ use.
 1311 78 A: The alloy is chemically identical to the certified reference.
 1312 79 B: Meeting that specification is a precondition for airframe use.
 1313 80 1. The response links A to the middle claim (it meets the specification), and the middle
 ↳ claim to B. It does not link A to B directly; that would be transitivity, which is not
 ↳ a relation to report.
 1314 81 2. Discourse sense: None.
 1315 82 3. Coupling: none.
 1316 83 4. [sense=None] [coupling=none] [evidence=none]
 1317 84
 1318 85 Your task:
 1319 86 RESPONSE: {{response}}
 87 A: {{atom_a}}
 88 B: {{atom_b}}

1321 A.3 PROMPT B, DEFAULT — SURROGATE-TOKEN CONDITIONAL STRENGTH

1322 2,182 characters. Placeholders: {{response}}, {{atom_a}}, {{atom_b}},
 1323 {{coupling}} (twice). This is the one prompt that does not end in a newline: it termi-
 1324 nates with Answer: including the trailing space, so that the model's next token is the surrogate
 1325 whose logprobs are read. Reproducing it without that trailing space changes the tokenization of the
 1326 answer position and therefore the estimate.
 1327

1328
 1329 1
 1330 2
 3 Instructions:
 1331 4 You are given the full RESPONSE, two claims A and B drawn from it, and a {{coupling}}
 ↳ relation that holds from A to B. Assuming A is TRUE, judge -- IN THE CONTEXT OF THIS
 1332 ↳ RESPONSE -- whether the relation's implication about B is credible: at least plausible
 1333 ↳ / more likely than not, NOT whether it is certain.
 5
 1334 6 - entailment or equivalence: given A and how the response uses it, is B at least plausibly
 ↳ TRUE (more likely than not)?
 1335 7 - contradiction or exclusive: given A and how the response uses it, is B at least plausibly
 ↳ FALSE (more likely than not)? (For "exclusive", A and B are exhaustive alternatives, so
 1336 ↳ A being true makes B false.)
 1337 8 - co_necessity: A and B are a pair of which at least one holds. Given the response rules
 ↳ out "neither", is it at least plausible that B holds when A does NOT?
 1338 9
 1339 10 Answer with a SINGLE WORD, the very first word of your reply: Yes or No.
 1340 11 - Answer "Yes" if the implication is credible/plausible (even if not certain).
 1341 12 - Answer "No" only if the implication is implausible or the response does not actually
 ↳ support it.
 1342 13 Do not output anything before the word Yes or No.
 1343 14
 1344 15 Example 1 (response supports a near-certain entailment):
 16 RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it
 ↳ meets the certified reference specification.
 1345 17 A: The new alloy is chemically identical to the certified reference alloy.
 1346 18 B: The new alloy meets the certified reference specification.
 1347 19 coupling: entailment
 1348 20 Answer: Yes
 1349 21
 22 Example 2 (weak but plausible, and the response draws the link -- still Yes):
 23 RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15
 ↳ percent over the same period.

1350 24 A: The company launched a flawed product last quarter.
 1351 25 B: The company's stock price fell 15 percent last quarter.
 1352 26 coupling: entailment
 1353 27 Answer: Yes
 1354 28
 1355 29 Example 3 (clear contradiction stated by the response):
 1356 30 RESPONSE: The official statement said no one was harmed, but three people died in the
 1357 31 incident.
 1358 32 A: No one was harmed in the incident.
 1359 33 B: Three people died in the incident.
 1360 34 coupling: contradiction
 1361 35 Answer: Yes
 1362 36
 1363 37 Your task:
 1364 38 RESPONSE: {{response}}
 1365 39 A: {{atom_a}}
 1366 40 B: {{atom_b}}
 1367 41 coupling: {{coupling}}
 1368 42 Answer:

A.4 PROMPT B, BASELINE — VERBALIZED CONDITIONAL STRENGTH

2,112 characters. Retained only for the comparison reported in §8; verbalized confidence is weakly calibrated (Xiong et al., 2024).

1369 1
 1370 2
 1371 3 Instructions:
 1372 4 You are given the full RESPONSE and two claims A and B drawn from it. Assume claim A is
 1373 5
 1374 6
 1375 7
 1376 8
 1377 9
 1378 10 Answer with a single probability in [0, 1] to two decimals, after one short sentence of
 1379 11 justification. A near-certain link is close to 1.00; a merely plausible link is around
 1380 12 0.60-0.70. End your answer with the probability wrapped in brackets on its own, exactly
 1381 13 in the form: [p=0.NN]
 1382 14
 1383 15 Example 1:
 1384 16 RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it
 1385 17 meets the certified reference specification.
 1386 18 A: The new alloy is chemically identical to the certified reference alloy.
 1387 19 B: The new alloy meets the certified reference specification.
 1388 20 coupling: entailment
 1389 21 Justification: chemical identity to a certified reference almost guarantees the
 1390 22 specification is met, so B follows very strongly from A.
 1391 23 [p=0.95]
 1392 24
 1393 25 Example 2:
 1394 26 RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15
 1395 27 percent over the same period.
 1396 28 A: The company launched a flawed product last quarter.
 1397 29 B: The company's stock price fell 15 percent last quarter.
 1398 30 coupling: entailment
 1399 31 Justification: a flawed product can plausibly drive a stock decline, but many other factors
 1400 32 affect price, so the link is only moderate.
 1401 33 [p=0.65]
 1402 34
 1403 35 Example 3:
 1404 36 RESPONSE: The official statement said no one was harmed, but three people died in the
 1405 37 incident.
 1406 38 A: No one was harmed in the incident.
 1407 39 B: Three people died in the incident.
 1408 40 coupling: contradiction
 1409 41 Justification: if no one was harmed, it is almost certain that B (three deaths) is false.
 1410 42 [p=0.93]
 1411 43
 1412 44 Your task:
 1413 45 RESPONSE: {{response}}
 1414 46 A: {{atom_a}}
 1415 47 B: {{atom_b}}
 1416 48 coupling: {{coupling}}

A.5 THE TWO SENSE MENUS

The full menu offers all twelve senses of Table 2 plus NONE:

Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast, $\hookrightarrow$ Concession, Alternative, Disjunction, Precedence, Succession, None
--

The restricted menu offers only the senses the evaluation corpus labels, which is the admissibility filter of §4.5 applied at elicitation time rather than after it:

Cause-Effect, Effect-Cause, Evidence, Restatement, Contrast, Concession, Alternative, $\hookrightarrow$ Disjunction, Precedence, None
--

Note that the restricted menu omits **Succession** as well as **Condition** and **Instantiation**, leaving nine relation-bearing senses. **Succession** is dropped because it is the mirror of **Precedence** and both compile to NONE, so offering both invites an arbitrary choice on a relation that produces no factor either way. The second-generation Prompt A’s body is written against this restricted menu; when it is run with the full menu, three senses are offered with no gloss and no coupling mapping in the prompt body, and we report which menu each arm used.

B THE REIFIED COHERENCE NODE, WORKED

To make Eq. 7 concrete, take the three-claim subnet consisting of the unresolved casualty conflict of Example 2 together with one entailment feeding it: claims a_7, a_{10} joined by a conflict at $p = 0.93$, and $a_4 \rightarrow a_7$ an entailment at $p = 0.65$, all priors $\frac{1}{2}$, with $\rho = \frac{1}{2}$. Writing out the eight-cell vote factor for each relation from Eq. 7 and enumerating the 2^4 worlds of (a_4, a_7, a_{10}, R) gives $\mathbb{P}(R=1) = 0.453$: below its prior, because both relations have violating worlds that carry appreciable mass, and the conflict’s violating world $(1, 1)$ carries the most.

Two properties of the construction are worth noting. The $R = 0$ branch is flat, so $R = 0$ asserts nothing and the node cannot be driven up merely by adding relations that happen to be satisfied. And the factor is a *noisy* AND rather than a hard one: a relation with $p_r < 1$ leaves some mass on its violating worlds even when $R = 1$, so a single strongly-violated relation reduces $\mathbb{P}(R=1)$ without pinning it to zero. The reason we do not adopt the readout despite this reasonable behaviour is Table 3: it dips at the concession variant for the reason of Prop. 3, and it requires the free parameter ρ , violating R4.

C CORRESPONDENCE WITH MARKOV LOGIC

The coherence MRF has a natural first-order encoding (Richardson & Domingos, 2006), with $\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \exp(\sum_k w_k n_k(\mathbf{a}))$ for $n_k(\mathbf{a})$ the number of true groundings of formula F_k . Evidence predicates *Entail*(i, j), *Contradict*(i, j), *Equiv*(i, j) and *Resolves*(h, i, j) carry the mined graph; *Holds*(i) is the query predicate. Working out the correspondence is instructive in both directions.

C.1 THE NAIVE ENCODING IS A DIFFERENT MODEL

The obvious encoding gives each relation one weighted clause, e.g. $\text{Entail}(i, j) \wedge a_i \Rightarrow a_j$ with $w = \text{logit}(p)$. This is *not* equivalent to Definition 1. Its induced pairwise factor is $[e^w, e^w, 1, e^w]$, which is not row-stochastic in the sense of Prop. 1: it rewards the vacuous $a_s = 0$ rows with e^w on *both* target states, so a clause is best satisfied by making its antecedent false. Brute force over the 2^{16} worlds of the running example gives mean marginal support 0.455 under the naive MLN against 0.587 under the corresponding MRF, with per-claim marginals differing by up to 0.31. These are two different models, not two encodings of one.

The failure is sharpest at the mode. The all-false world satisfies every implication vacuously, so under the naive encoding the most probable configuration of any purely-implicational graph is the one in which the response asserts nothing — the maximum-a-posteriori reading of a response is that none of it holds. This is Prop. 2 appearing in a second guise, and it is the same lesson: an encoding that scores rule satisfaction rather than belief can be maximized by having nothing to satisfy.

C.2 A THREE-CLAUSE EXPANSION IS EXACTLY EQUIVALENT

Writing the log-potential in the general pairwise form $\ln \psi(a_s, a_t) = \alpha + \beta a_s + \gamma a_t + \delta a_s a_t$ and matching it against the with-priors tables of Table 1 gives, for π_s the source prior:

τ	α (unit)	β (source)	γ (target)	δ (conjunction)
ENTAILMENT	$\ln \frac{1-p}{\pi_s}$	0	0	$+\text{logit}(p)$
CONTRADICTION	$\ln \frac{p}{\pi_s}$	0	0	$-\text{logit}(p)$
EQUIVALENCE	$\ln \frac{1-p}{p}$	$\ln \frac{1-p}{p}$	0	$+2 \text{logit}(p)$

The conjunction weight is exactly $\pm \text{logit}(p)$ — the interaction the implication was meant to carry all along — and the unit clause is the correction that restores row-stochasticity. With this expansion the ground Markov logic network reproduces the MRF exactly: mean support 0.5866 and $\log Z = -9.751$ for both, with zero per-claim difference to machine precision.

We establish the equivalence for the three original couplings only. EXCLUSIVE and CO_NECESSITY are not tabulated here, so the claim “the MLN encoding reproduces the MRF” should be read as scoped to the three-coupling fragment, which is a real limitation given that both worked examples of §9 use EXCLUSIVE conflicts.

C.3 WHICH VIEW TO KEEP, AND WHY BOTH

The reason to prefer the MRF for the measure itself is Remark 1: its pairwise factor family is closed at three parameters, so the inferential vocabulary cannot quietly grow as senses are added, and every relation type is guaranteed to be a 2×2 potential. The reason to keep the Markov logic view is that the rules our model cannot express are naturally written there — a transitivity rule $\text{Entail}(i, j) \wedge \text{Entail}(j, k) \Rightarrow \text{Entail}(i, k)$, a concession-cancellation rule $\text{Resolves}(h, i, j) \wedge a_h \Rightarrow \neg \text{Penalize}(i, j)$ replacing the fixed discount of Eq. 3, and a double-conflict rule — and their weights could be *learned* from labelled ladders rather than set by hand. That is the route past the two limitations §10 names, and it requires an MLN solver (MC-SAT for marginals, MaxWalkSAT for the mode) rather than the bucket-elimination stack used here.

D SERIALIZATION AND THE VARIABLE-ORDERING TRAP

One implementation detail is worth recording because it is silent when wrong. Networks are serialized in UAI format, whose variables are identified by integer index; the index order is the sort order of (cardinality, name), and with uniformly binary claim variables that reduces to *lexicographic order by name*. So a_{10} sorts before a_2 . Inference returns marginals by index, and mapping them back by any other convention — source order, numeric order — silently permutes marginals across claims. The score is unchanged, since Eq. 5 is a mean, but every per-claim diagnostic is then attributed to the wrong claim, and the “claim dragged below its prior” readout points at an innocent claim. The auxiliary variables of §5.2 are named with a prefix that sorts after all claim names for the same reason, so that adding them cannot shift the claim indices.